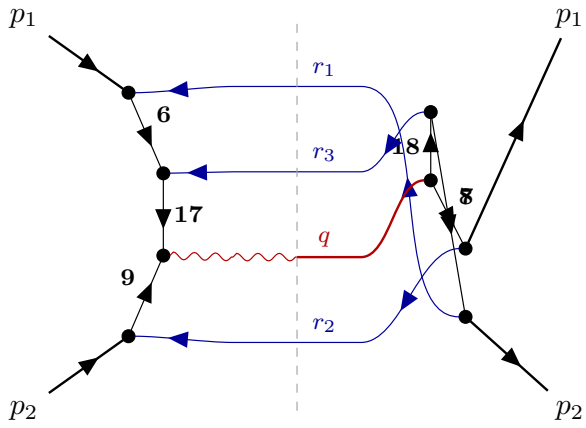
Integral families as cut diagrams

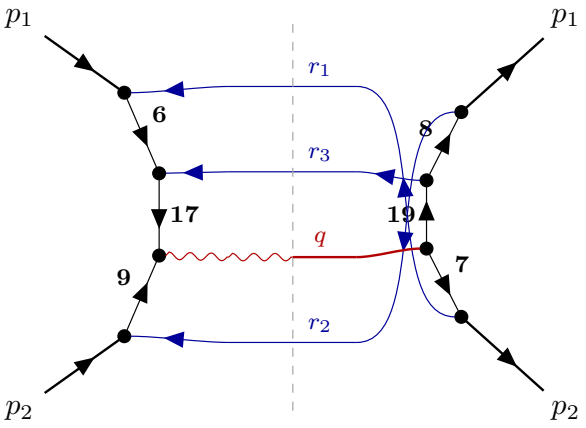
Amplitude (left) × conjugate (right) for $p_1p_2 \to V(q)\, r_1r_2r_3$, glued along the cut (dashed): r_1, r_2, r_3 (blue), q (red). The rapidity condition $-q\cdot p_1 + u\, q\cdot p_2 = 0$ of every family is not a line: the q line is drawn wavy up to the cut ($q^2 = m^2$) and as a double line after it (the rapidity condition). Internal lines carry their DD number. Arrows: momentum flow with all legs incoming ($p_1 + p_2 + r_1 + r_2 + r_3 + q = 0$): in the amplitude every line carries the sum of the incoming momenta on its far side, e.g. line 6 carries $p_1 + r_1$; the conjugate is the mirror image, so p_1, p_2 leave on the right and r_i, q flow through the cut from right to left (the physical final-state momenta $-r_i$ from left to right). **Regular propagators** (DD numbers, all legs incoming): $6 = (p_1 + r_1)^2$; $7 = (p_2 + r_1)^2$; $8 = (p_1 + r_2)^2$; $9 = (p_2 + r_2)^2$; $10 = (p_1 + r_3)^2$; $11 = (p_2 + r_3)^2$; $12 = (r_1 + r_2)^2$; $13 = (r_2 + r_3)^2$; $14 = (r_1 + r_3)^2$; $15 = (p_1 + r_1 + r_2)^2$; $16 = (p_2 + r_1 + r_2)^2$; $17 = (p_1 + r_1 + r_3)^2$; $18 = (p_2 + r_1 + r_3)^2$; $19 = (p_1 + r_2 + r_3)^2$; $20 = (p_2 + r_2 + r_3)^2$; $21 = (r_1 + r_2 + r_3)^2$; $22 = (p_1 + r_1 + r_2 + r_3)^2$; $23 = (p_2 + r_1 + r_2 + r_3)^2$; $24 = (p_1 + p_2 + r_1)^2$; $25 = (p_1 + p_2 + r_2)^2$; $26 = (p_1 + p_2 + r_3)^2$; $27 = (p_1 + p_2 + r_1 + r_2)^2$; $28 = (p_1 + p_2 + r_1 + r_3)^2$; $29 = (p_1 + p_2 + r_2 + r_3)^2$.



FAM[1] (topo1)

6	$(p_1 + r_1)^2$	amplitude
9	$(p_2 + r_2)^2$	amplitude
17	$(p_1 + r_1 + r_3)^2$	amplitude
7	$(p_2 + r_1)^2$	conjugate
8	$(p_1 + r_2)^2$	conjugate
18	$(p_2 + r_1 + r_3)^2$	conjugate

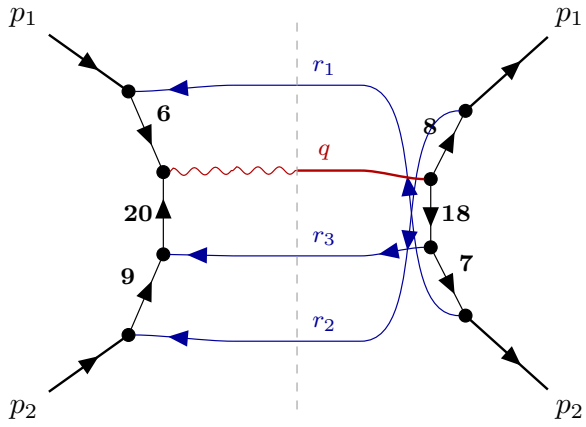
irreducible numerators: $27 = (p_1 + p_2 + r_1 + r_2)^2$



FAM[7] (topo7)

6	$(p_1 + r_1)^2$	amplitude
9	$(p_2 + r_2)^2$	amplitude
17	$(p_1 + r_1 + r_3)^2$	amplitude
7	$(p_2 + r_1)^2$	conjugate
8	$(p_1 + r_2)^2$	conjugate
19	$(p_1 + r_2 + r_3)^2$	conjugate

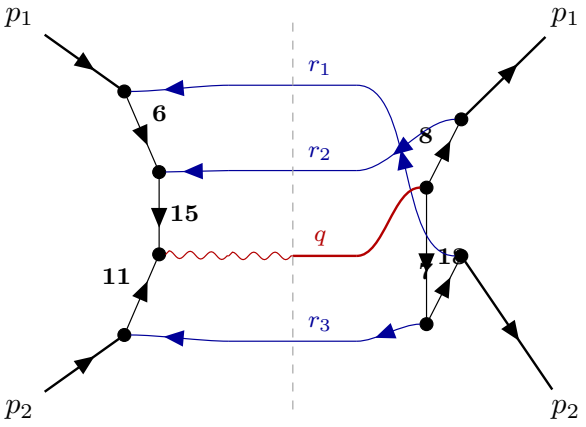
irreducible numerators: $13 = (r_2 + r_3)^2$



FAM[13] (topo13)

6	$(p_1 + r_1)^2$	amplitude
9	$(p_2 + r_2)^2$	amplitude
20	$(p_2 + r_2 + r_3)^2$	amplitude
7	$(p_2 + r_1)^2$	conjugate
8	$(p_1 + r_2)^2$	conjugate
18	$(p_2 + r_1 + r_3)^2$	conjugate

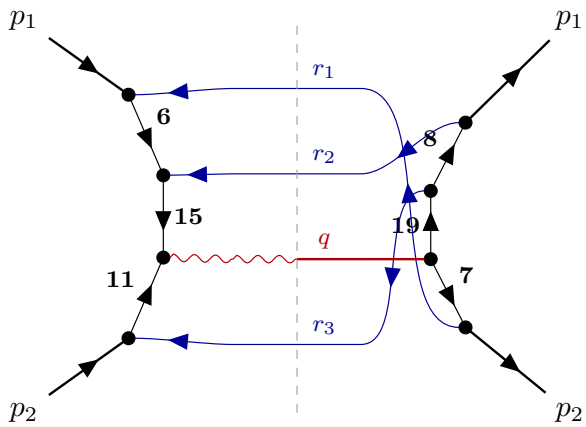
irreducible numerators: $13 = (r_2 + r_3)^2$



FAM[19] (topo19)

6	$(p_1 + r_1)^2$	amplitude
11	$(p_2 + r_3)^2$	amplitude
15	$(p_1 + r_1 + r_2)^2$	amplitude
7	$(p_2 + r_1)^2$	conjugate
8	$(p_1 + r_2)^2$	conjugate
18	$(p_2 + r_1 + r_3)^2$	conjugate

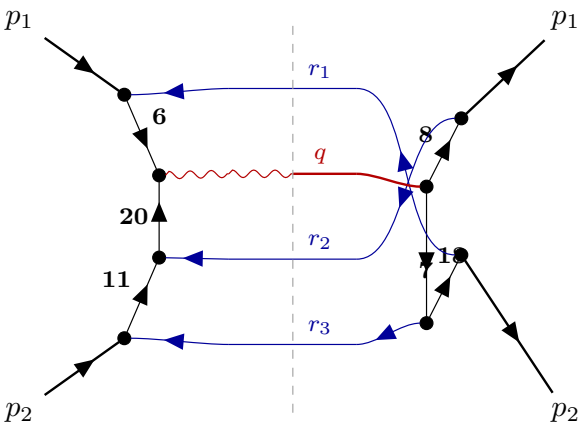
irreducible numerators: $27 = (p_1 + p_2 + r_1 + r_2)^2$



FAM[25] (topo25)

6	$(p_1 + r_1)^2$	amplitude
11	$(p_2 + r_3)^2$	amplitude
15	$(p_1 + r_1 + r_2)^2$	amplitude
7	$(p_2 + r_1)^2$	conjugate
8	$(p_1 + r_2)^2$	conjugate
19	$(p_1 + r_2 + r_3)^2$	conjugate

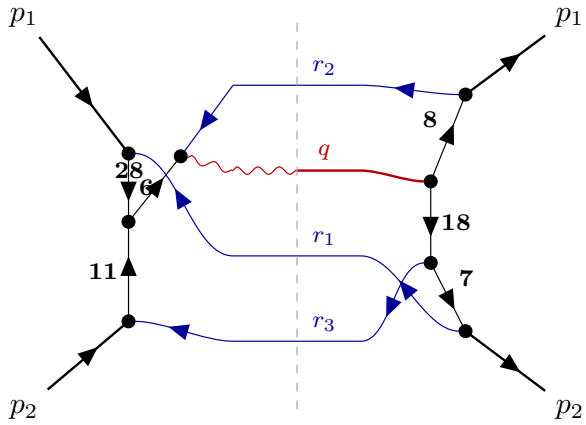
irreducible numerators: $22 = (p_1 + r_1 + r_2 + r_3)^2$



FAM[31] (topo31)

6	$(p_1 + r_1)^2$	amplitude
11	$(p_2 + r_3)^2$	amplitude
20	$(p_2 + r_2 + r_3)^2$	amplitude
7	$(p_2 + r_1)^2$	conjugate
8	$(p_1 + r_2)^2$	conjugate
18	$(p_2 + r_1 + r_3)^2$	conjugate

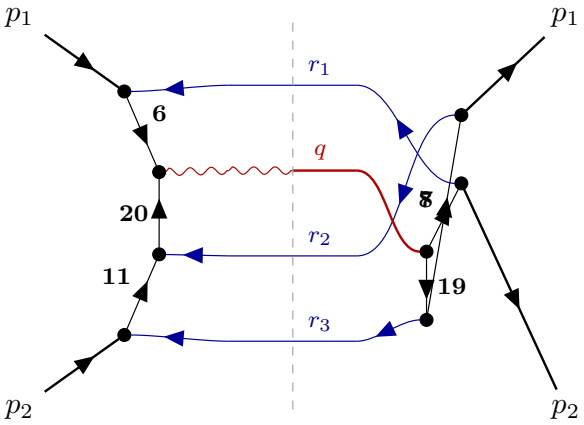
irreducible numerators: $23 = (p_2 + r_1 + r_2 + r_3)^2$



FAM[37] (topo37)

6	$(p_1 + r_1)^2$	amplitude
11	$(p_2 + r_3)^2$	amplitude
28	$(p_1 + p_2 + r_1 + r_3)^2$	amplitude
7	$(p_2 + r_1)^2$	conjugate
8	$(p_1 + r_2)^2$	conjugate
18	$(p_2 + r_1 + r_3)^2$	conjugate

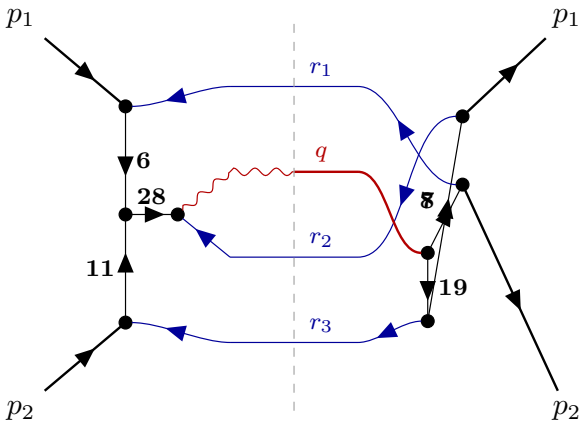
irreducible numerators: $27 = (p_1 + p_2 + r_1 + r_2)^2$



FAM[43] (topo43)

6	$(p_1 + r_1)^2$	amplitude
11	$(p_2 + r_3)^2$	amplitude
20	$(p_2 + r_2 + r_3)^2$	amplitude
7	$(p_2 + r_1)^2$	conjugate
8	$(p_1 + r_2)^2$	conjugate
19	$(p_1 + r_2 + r_3)^2$	conjugate

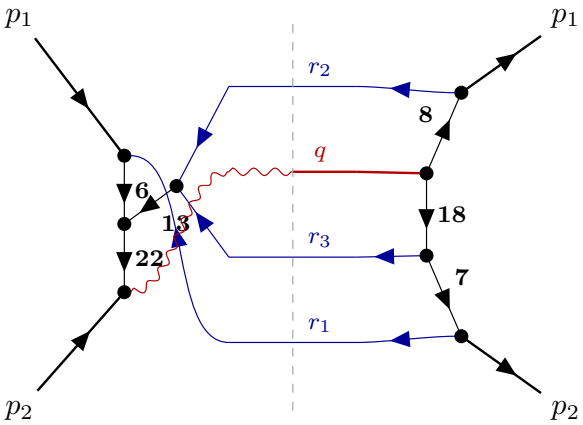
irreducible numerators: $27 = (p_1 + p_2 + r_1 + r_2)^2$



FAM[49] (topo49)

6	$(p_1 + r_1)^2$	amplitude
11	$(p_2 + r_3)^2$	amplitude
28	$(p_1 + p_2 + r_1 + r_3)^2$	amplitude
7	$(p_2 + r_1)^2$	conjugate
8	$(p_1 + r_2)^2$	conjugate
19	$(p_1 + r_2 + r_3)^2$	conjugate

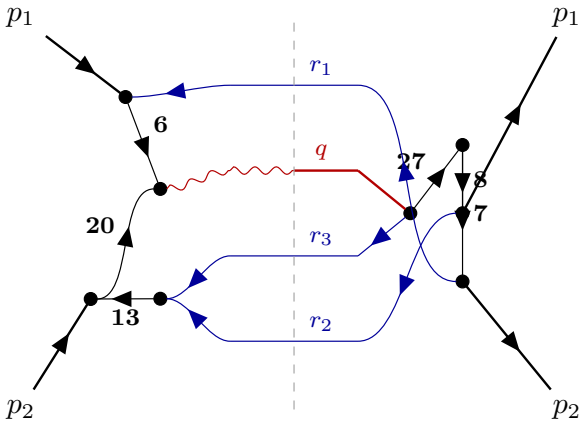
irreducible numerators: $13 = (r_2 + r_3)^2$



FAM[55] (topo55)

6	$(p_1 + r_1)^2$	amplitude
13	$(r_2 + r_3)^2$	amplitude
22	$(p_1 + r_1 + r_2 + r_3)^2$	amplitude
7	$(p_2 + r_1)^2$	conjugate
8	$(p_1 + r_2)^2$	conjugate
18	$(p_2 + r_1 + r_3)^2$	conjugate

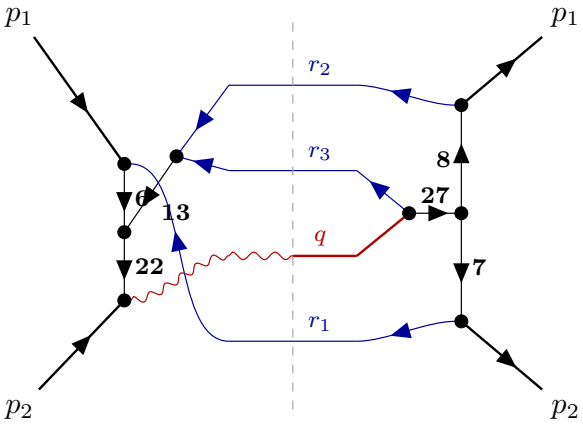
irreducible numerators: $14 = (r_1 + r_3)^2$



FAM[61] (topo61)

6	$(p_1 + r_1)^2$	amplitude
13	$(r_2 + r_3)^2$	amplitude
20	$(p_2 + r_2 + r_3)^2$	amplitude
7	$(p_2 + r_1)^2$	conjugate
8	$(p_1 + r_2)^2$	conjugate
27	$(p_1 + p_2 + r_1 + r_2)^2$	conjugate

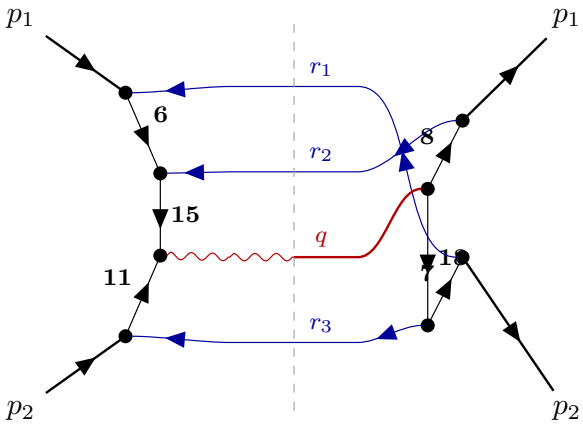
irreducible numerators: $25 = (p_1 + p_2 + r_2)^2$



FAM[67] (topo67)

6	$(p_1 + r_1)^2$	amplitude
13	$(r_2 + r_3)^2$	amplitude
22	$(p_1 + r_1 + r_2 + r_3)^2$	amplitude
7	$(p_2 + r_1)^2$	conjugate
8	$(p_1 + r_2)^2$	conjugate
27	$(p_1 + p_2 + r_1 + r_2)^2$	conjugate

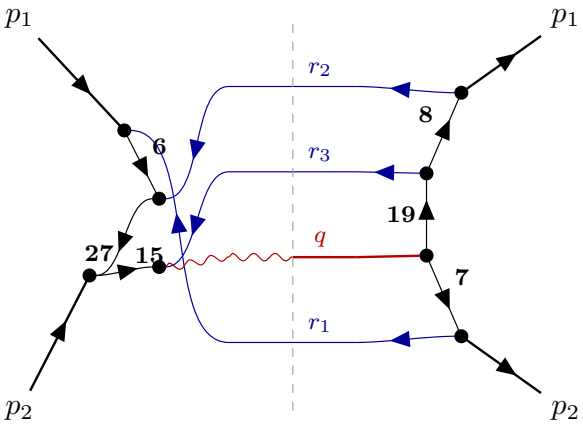
irreducible numerators: $15 = (p_1 + r_1 + r_2)^2$



FAM[73] (topo73)

6	$(p_1 + r_1)^2$	amplitude
11	$(p_2 + r_3)^2$	amplitude
15	$(p_1 + r_1 + r_2)^2$	amplitude
7	$(p_2 + r_1)^2$	conjugate
8	$(p_1 + r_2)^2$	conjugate
18	$(p_2 + r_1 + r_3)^2$	conjugate

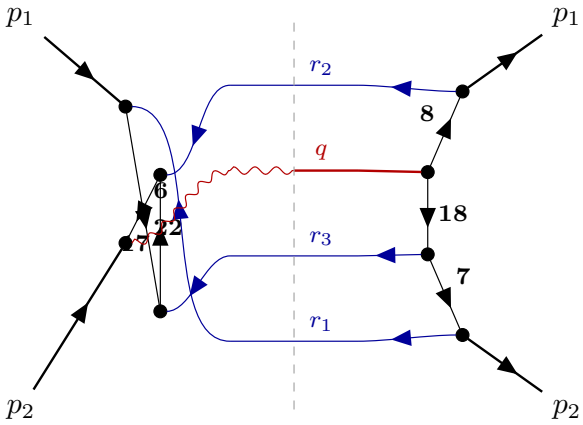
irreducible numerators: $22 = (p_1 + r_1 + r_2 + r_3)^2$



FAM[79] (topo79)

6	$(p_1 + r_1)^2$	amplitude
15	$(p_1 + r_1 + r_2)^2$	amplitude
27	$(p_1 + p_2 + r_1 + r_2)^2$	amplitude
7	$(p_2 + r_1)^2$	conjugate
8	$(p_1 + r_2)^2$	conjugate
19	$(p_1 + r_2 + r_3)^2$	conjugate

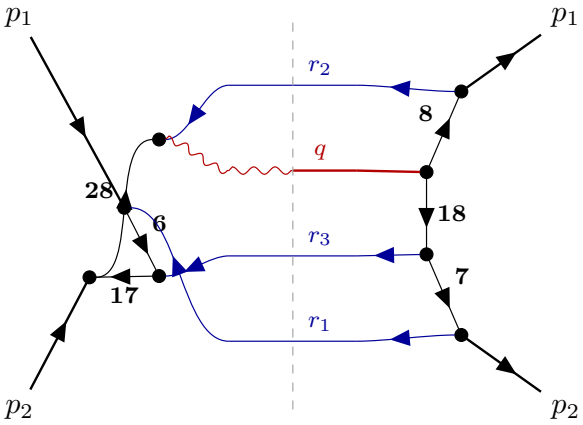
irreducible numerators: $13 = (r_2 + r_3)^2$



FAM[85] (topo85)

6	$(p_1 + r_1)^2$	amplitude
17	$(p_1 + r_1 + r_3)^2$	amplitude
22	$(p_1 + r_1 + r_2 + r_3)^2$	amplitude
7	$(p_2 + r_1)^2$	conjugate
8	$(p_1 + r_2)^2$	conjugate
18	$(p_2 + r_1 + r_3)^2$	conjugate

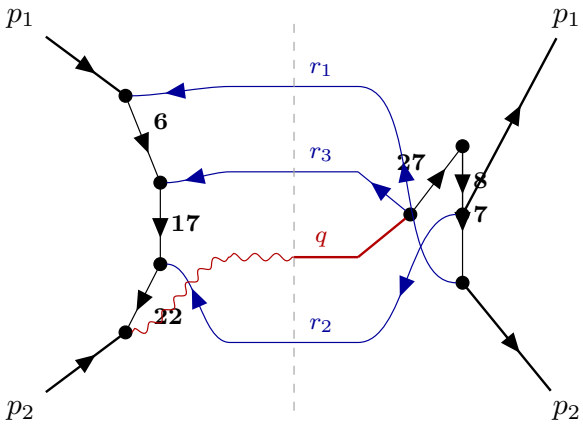
irreducible numerators: $19 = (p_1 + r_2 + r_3)^2$



FAM[91] (topo91)

6	$(p_1 + r_1)^2$	amplitude
17	$(p_1 + r_1 + r_3)^2$	amplitude
28	$(p_1 + p_2 + r_1 + r_3)^2$	amplitude
7	$(p_2 + r_1)^2$	conjugate
8	$(p_1 + r_2)^2$	conjugate
18	$(p_2 + r_1 + r_3)^2$	conjugate

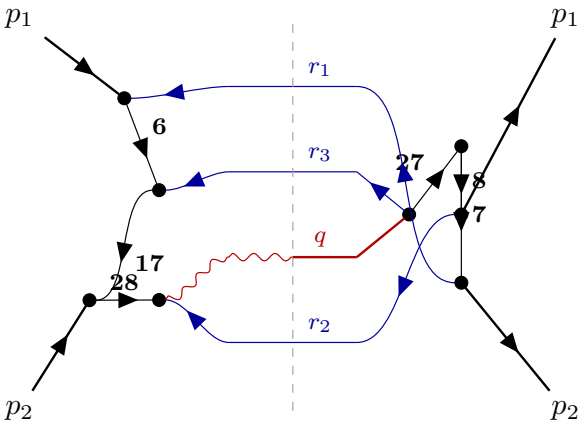
irreducible numerators: $19 = (p_1 + r_2 + r_3)^2$



FAM[97] (topo97)

6	$(p_1 + r_1)^2$	amplitude
17	$(p_1 + r_1 + r_3)^2$	amplitude
22	$(p_1 + r_1 + r_2 + r_3)^2$	amplitude
7	$(p_2 + r_1)^2$	conjugate
8	$(p_1 + r_2)^2$	conjugate
27	$(p_1 + p_2 + r_1 + r_2)^2$	conjugate

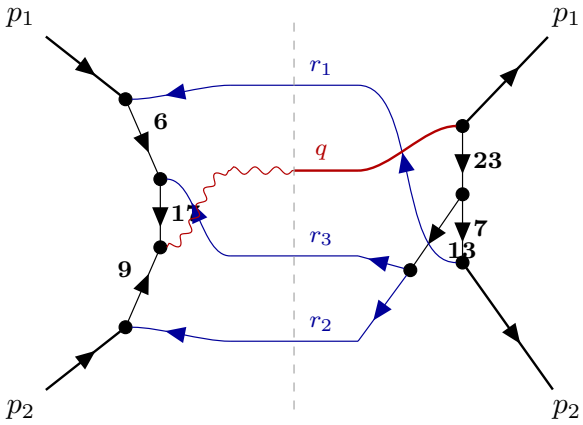
irreducible numerators: $25 = (p_1 + p_2 + r_2)^2$



FAM[103] (topo103)

6	$(p_1 + r_1)^2$	amplitude
17	$(p_1 + r_1 + r_3)^2$	amplitude
28	$(p_1 + p_2 + r_1 + r_3)^2$	amplitude
7	$(p_2 + r_1)^2$	conjugate
8	$(p_1 + r_2)^2$	conjugate
27	$(p_1 + p_2 + r_1 + r_2)^2$	conjugate

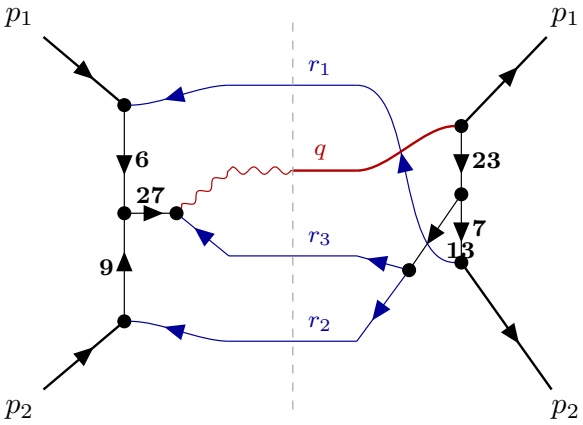
irreducible numerators: $14 = (r_1 + r_3)^2$



FAM[109] (topo109)

6	$(p_1 + r_1)^2$	amplitude
9	$(p_2 + r_2)^2$	amplitude
17	$(p_1 + r_1 + r_3)^2$	amplitude
7	$(p_2 + r_1)^2$	conjugate
13	$(r_2 + r_3)^2$	conjugate
23	$(p_2 + r_1 + r_2 + r_3)^2$	conjugate

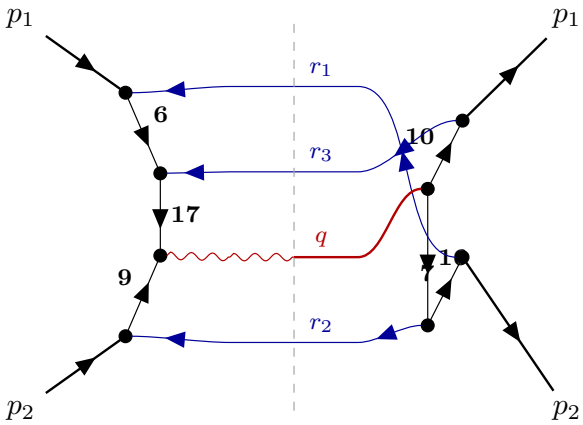
irreducible numerators: $14 = (r_1 + r_3)^2$



FAM[115] (topo115)

6	$(p_1 + r_1)^2$	amplitude
9	$(p_2 + r_2)^2$	amplitude
27	$(p_1 + p_2 + r_1 + r_2)^2$	amplitude
7	$(p_2 + r_1)^2$	conjugate
13	$(r_2 + r_3)^2$	conjugate
23	$(p_2 + r_1 + r_2 + r_3)^2$	conjugate

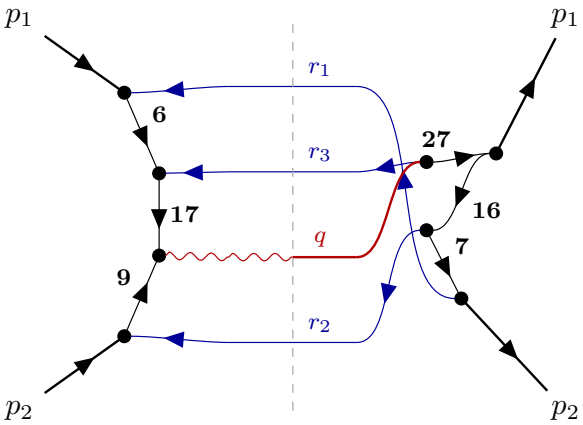
irreducible numerators: $16 = (p_2 + r_1 + r_2)^2$



FAM[121] (topo121)

6	$(p_1 + r_1)^2$	amplitude
9	$(p_2 + r_2)^2$	amplitude
17	$(p_1 + r_1 + r_3)^2$	amplitude
7	$(p_2 + r_1)^2$	conjugate
10	$(p_1 + r_3)^2$	conjugate
16	$(p_2 + r_1 + r_2)^2$	conjugate

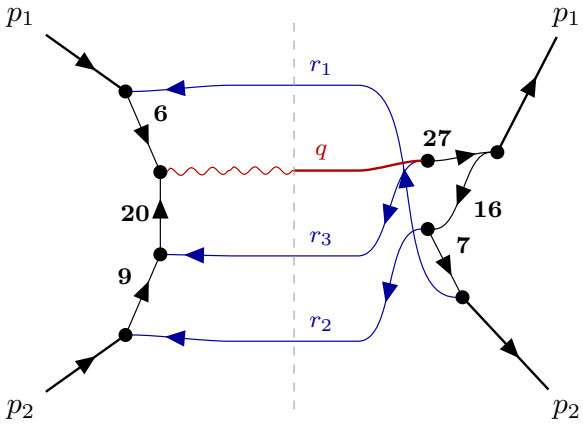
irreducible numerators: $23 = (p_2 + r_1 + r_2 + r_3)^2$



FAM[127] (topo127)

6	$(p_1 + r_1)^2$	amplitude
9	$(p_2 + r_2)^2$	amplitude
17	$(p_1 + r_1 + r_3)^2$	amplitude
7	$(p_2 + r_1)^2$	conjugate
16	$(p_2 + r_1 + r_2)^2$	conjugate
27	$(p_1 + p_2 + r_1 + r_2)^2$	conjugate

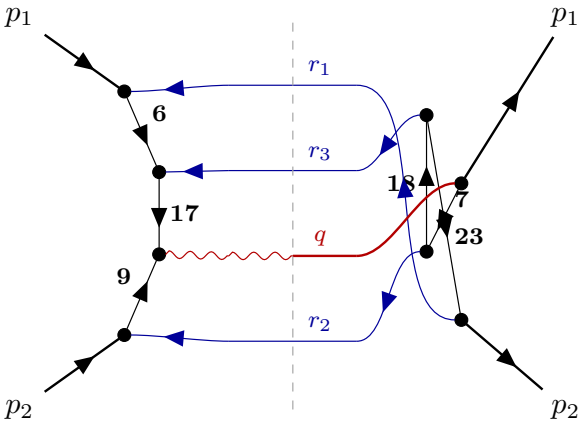
irreducible numerators: $22 = (p_1 + r_1 + r_2 + r_3)^2$



FAM[133] (topo133)

6	$(p_1 + r_1)^2$	amplitude
9	$(p_2 + r_2)^2$	amplitude
20	$(p_2 + r_2 + r_3)^2$	amplitude
7	$(p_2 + r_1)^2$	conjugate
16	$(p_2 + r_1 + r_2)^2$	conjugate
27	$(p_1 + p_2 + r_1 + r_2)^2$	conjugate

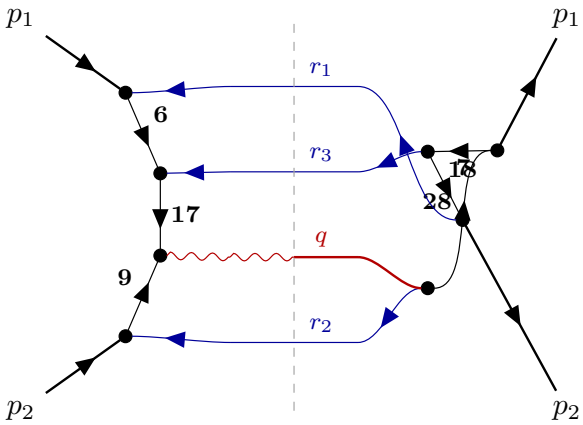
irreducible numerators: $13 = (r_2 + r_3)^2$



FAM[139] (topo139)

6	$(p_1 + r_1)^2$	amplitude
9	$(p_2 + r_2)^2$	amplitude
17	$(p_1 + r_1 + r_3)^2$	amplitude
7	$(p_2 + r_1)^2$	conjugate
18	$(p_2 + r_1 + r_3)^2$	conjugate
23	$(p_2 + r_1 + r_2 + r_3)^2$	conjugate

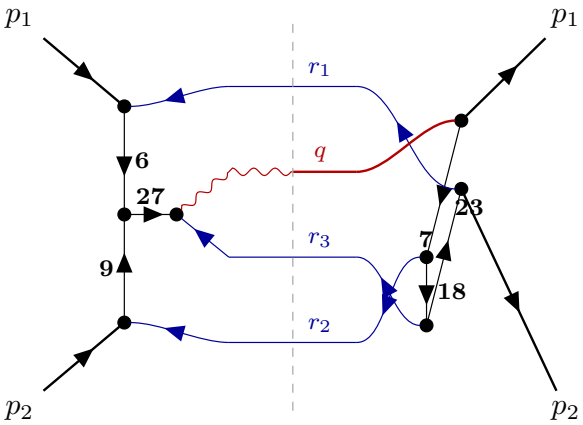
irreducible numerators: $20 = (p_2 + r_2 + r_3)^2$



FAM[145] (topo145)

6	$(p_1 + r_1)^2$	amplitude
9	$(p_2 + r_2)^2$	amplitude
17	$(p_1 + r_1 + r_3)^2$	amplitude
7	$(p_2 + r_1)^2$	conjugate
18	$(p_2 + r_1 + r_3)^2$	conjugate
28	$(p_1 + p_2 + r_1 + r_3)^2$	conjugate

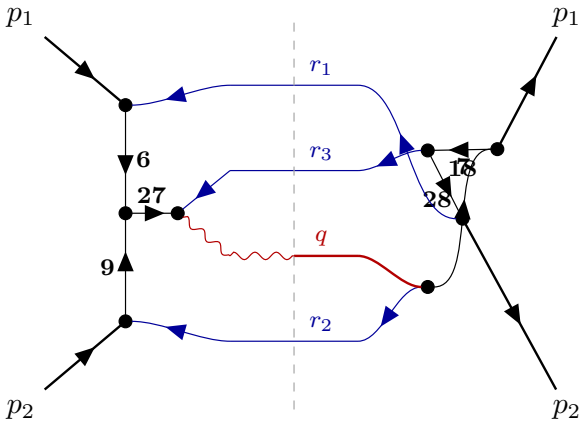
irreducible numerators: $20 = (p_2 + r_2 + r_3)^2$



FAM[151] (topo151)

6	$(p_1 + r_1)^2$	amplitude
9	$(p_2 + r_2)^2$	amplitude
27	$(p_1 + p_2 + r_1 + r_2)^2$	amplitude
7	$(p_2 + r_1)^2$	conjugate
18	$(p_2 + r_1 + r_3)^2$	conjugate
23	$(p_2 + r_1 + r_2 + r_3)^2$	conjugate

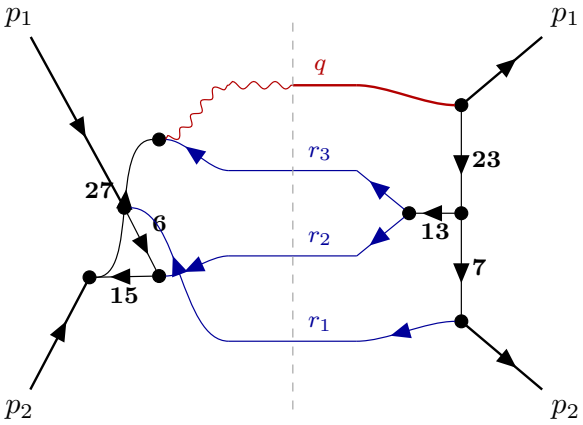
irreducible numerators: $25 = (p_1 + p_2 + r_2)^2$



FAM[157] (topo157)

6	$(p_1 + r_1)^2$	amplitude
9	$(p_2 + r_2)^2$	amplitude
27	$(p_1 + p_2 + r_1 + r_2)^2$	amplitude
7	$(p_2 + r_1)^2$	conjugate
18	$(p_2 + r_1 + r_3)^2$	conjugate
28	$(p_1 + p_2 + r_1 + r_3)^2$	conjugate

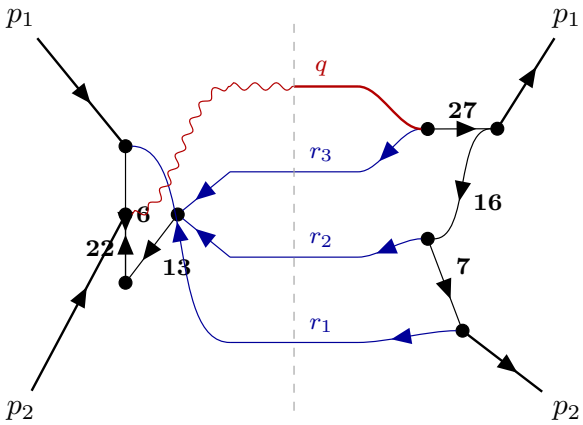
irreducible numerators: $14 = (r_1 + r_3)^2$



FAM[163] (topo163)

6	$(p_1 + r_1)^2$	amplitude
15	$(p_1 + r_1 + r_2)^2$	amplitude
27	$(p_1 + p_2 + r_1 + r_2)^2$	amplitude
7	$(p_2 + r_1)^2$	conjugate
13	$(r_2 + r_3)^2$	conjugate
23	$(p_2 + r_1 + r_2 + r_3)^2$	conjugate

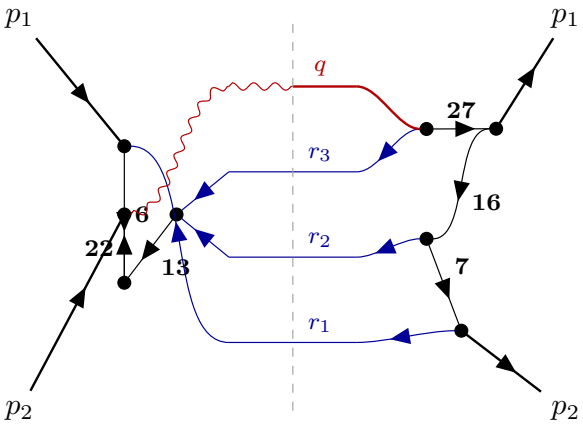
irreducible numerators: $12 = (r_1 + r_2)^2$



FAM[169] (topo169)

6	$(p_1 + r_1)^2$	amplitude
13	$(r_2 + r_3)^2$	amplitude
22	$(p_1 + r_1 + r_2 + r_3)^2$	amplitude
7	$(p_2 + r_1)^2$	conjugate
16	$(p_2 + r_1 + r_2)^2$	conjugate
27	$(p_1 + p_2 + r_1 + r_2)^2$	conjugate

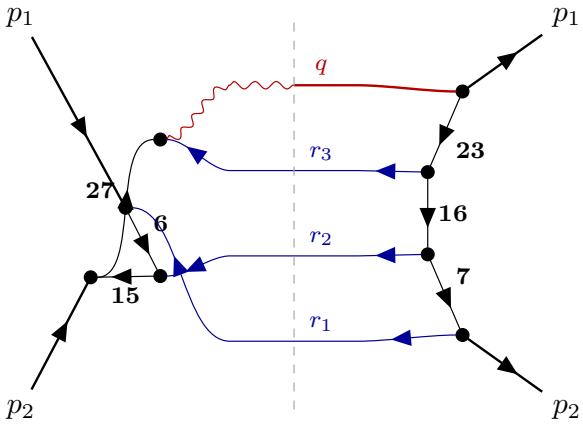
irreducible numerators: $12 = (r_1 + r_2)^2$



FAM[175] (topo175)

6	$(p_1 + r_1)^2$	amplitude
13	$(r_2 + r_3)^2$	amplitude
22	$(p_1 + r_1 + r_2 + r_3)^2$	amplitude
7	$(p_2 + r_1)^2$	conjugate
16	$(p_2 + r_1 + r_2)^2$	conjugate
27	$(p_1 + p_2 + r_1 + r_2)^2$	conjugate

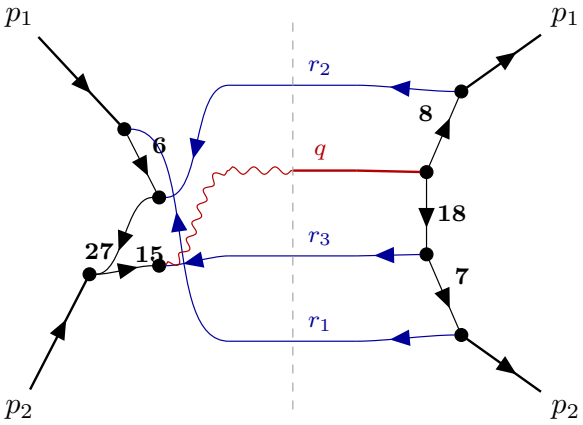
irreducible numerators: $15 = (p_1 + r_1 + r_2)^2$



FAM[181] (topo181)

6	$(p_1 + r_1)^2$	amplitude
15	$(p_1 + r_1 + r_2)^2$	amplitude
27	$(p_1 + p_2 + r_1 + r_2)^2$	amplitude
7	$(p_2 + r_1)^2$	conjugate
16	$(p_2 + r_1 + r_2)^2$	conjugate
23	$(p_2 + r_1 + r_2 + r_3)^2$	conjugate

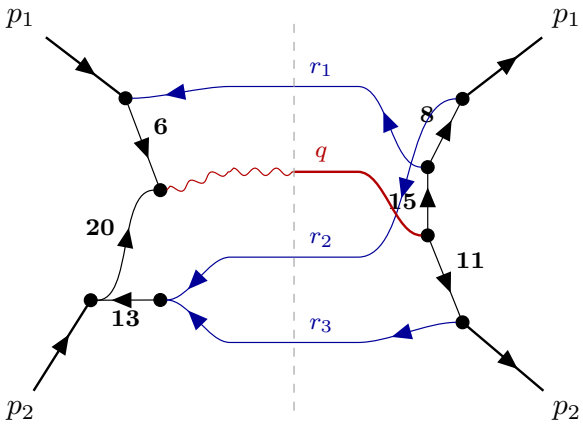
irreducible numerators: $18 = (p_2 + r_1 + r_3)^2$



FAM[187] (topo187)

6	$(p_1 + r_1)^2$	amplitude
15	$(p_1 + r_1 + r_2)^2$	amplitude
27	$(p_1 + p_2 + r_1 + r_2)^2$	amplitude
7	$(p_2 + r_1)^2$	conjugate
8	$(p_1 + r_2)^2$	conjugate
18	$(p_2 + r_1 + r_3)^2$	conjugate

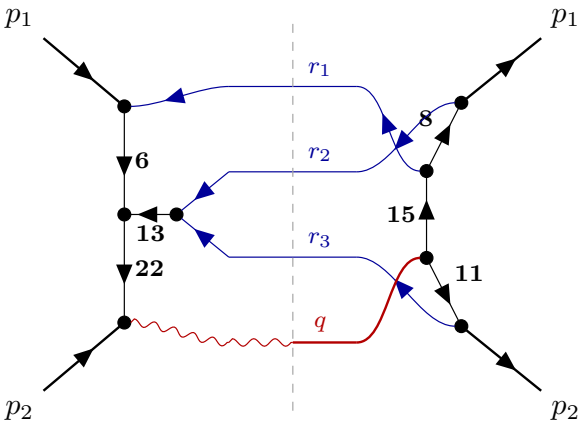
irreducible numerators: $28 = (p_1 + p_2 + r_1 + r_3)^2$



FAM[193] (topo193)

6	$(p_1 + r_1)^2$	amplitude
13	$(r_2 + r_3)^2$	amplitude
20	$(p_2 + r_2 + r_3)^2$	amplitude
8	$(p_1 + r_2)^2$	conjugate
11	$(p_2 + r_3)^2$	conjugate
15	$(p_1 + r_1 + r_2)^2$	conjugate

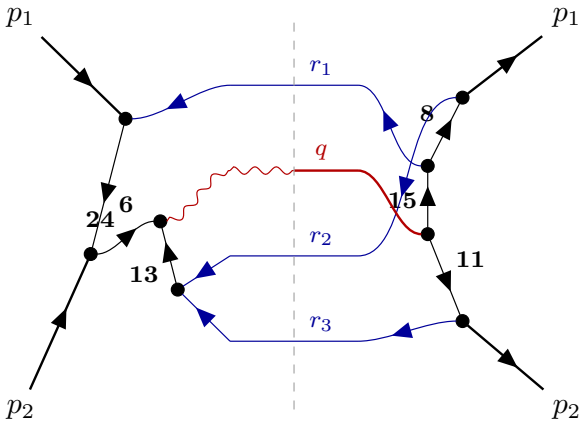
irreducible numerators: $21 = (r_1 + r_2 + r_3)^2$



FAM[199] (topo199)

6	$(p_1 + r_1)^2$	amplitude
13	$(r_2 + r_3)^2$	amplitude
22	$(p_1 + r_1 + r_2 + r_3)^2$	amplitude
8	$(p_1 + r_2)^2$	conjugate
11	$(p_2 + r_3)^2$	conjugate
15	$(p_1 + r_1 + r_2)^2$	conjugate

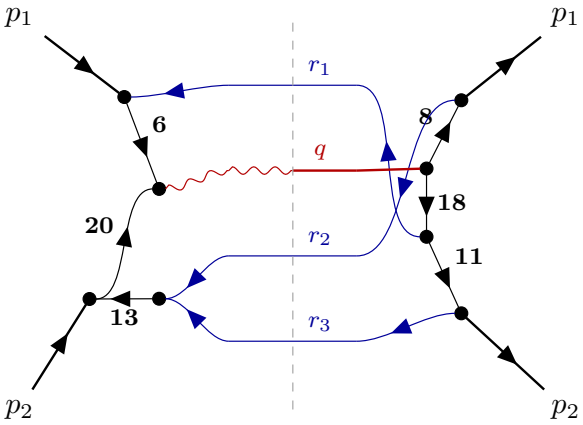
irreducible numerators: $29 = (p_1 + p_2 + r_2 + r_3)^2$



FAM[205] (topo205)

6	$(p_1 + r_1)^2$	amplitude
13	$(r_2 + r_3)^2$	amplitude
24	$(p_1 + p_2 + r_1)^2$	amplitude
8	$(p_1 + r_2)^2$	conjugate
11	$(p_2 + r_3)^2$	conjugate
15	$(p_1 + r_1 + r_2)^2$	conjugate

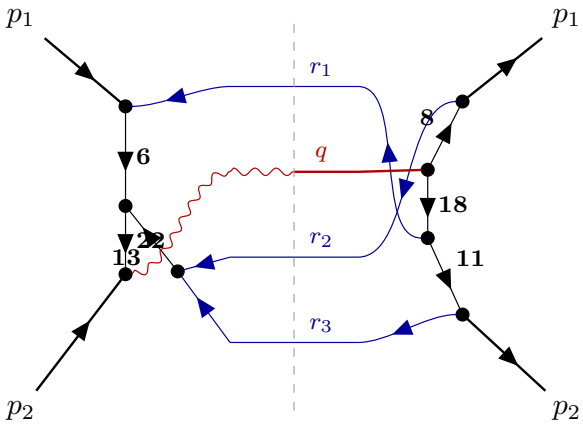
irreducible numerators: $27 = (p_1 + p_2 + r_1 + r_2)^2$



FAM[211] (topo211)

6	$(p_1 + r_1)^2$	amplitude
13	$(r_2 + r_3)^2$	amplitude
20	$(p_2 + r_2 + r_3)^2$	amplitude
8	$(p_1 + r_2)^2$	conjugate
11	$(p_2 + r_3)^2$	conjugate
18	$(p_2 + r_1 + r_3)^2$	conjugate

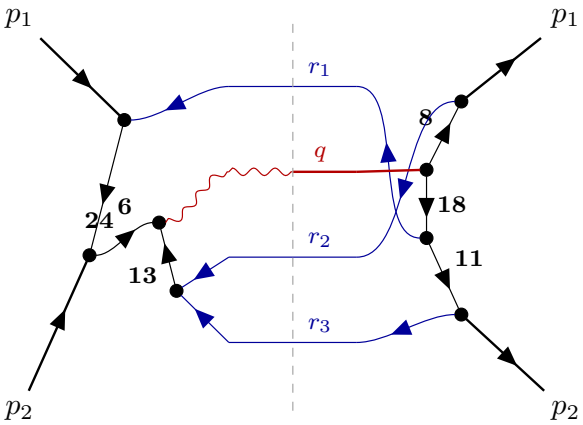
irreducible numerators: $19 = (p_1 + r_2 + r_3)^2$



FAM[217] (topo217)

6	$(p_1 + r_1)^2$	amplitude
13	$(r_2 + r_3)^2$	amplitude
22	$(p_1 + r_1 + r_2 + r_3)^2$	amplitude
8	$(p_1 + r_2)^2$	conjugate
11	$(p_2 + r_3)^2$	conjugate
18	$(p_2 + r_1 + r_3)^2$	conjugate

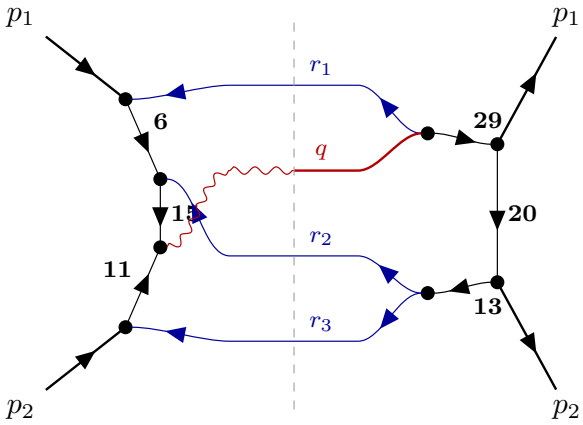
irreducible numerators: $17 = (p_1 + r_1 + r_3)^2$



FAM[223] (topo223)

6	$(p_1 + r_1)^2$	amplitude
13	$(r_2 + r_3)^2$	amplitude
24	$(p_1 + p_2 + r_1)^2$	amplitude
8	$(p_1 + r_2)^2$	conjugate
11	$(p_2 + r_3)^2$	conjugate
18	$(p_2 + r_1 + r_3)^2$	conjugate

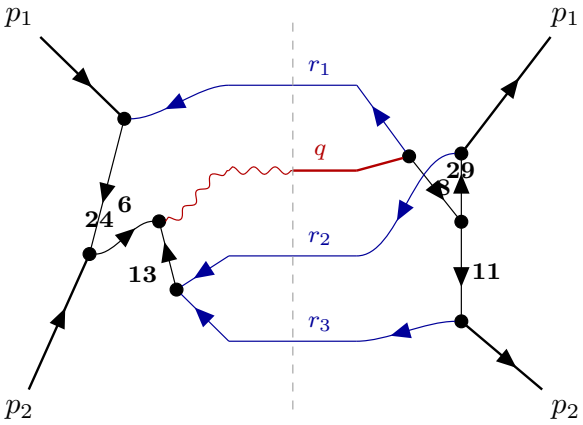
irreducible numerators: $28 = (p_1 + p_2 + r_1 + r_3)^2$



FAM[229] (topo229)

6	$(p_1 + r_1)^2$	amplitude
11	$(p_2 + r_3)^2$	amplitude
15	$(p_1 + r_1 + r_2)^2$	amplitude
13	$(r_2 + r_3)^2$	conjugate
20	$(p_2 + r_2 + r_3)^2$	conjugate
29	$(p_1 + p_2 + r_2 + r_3)^2$	conjugate

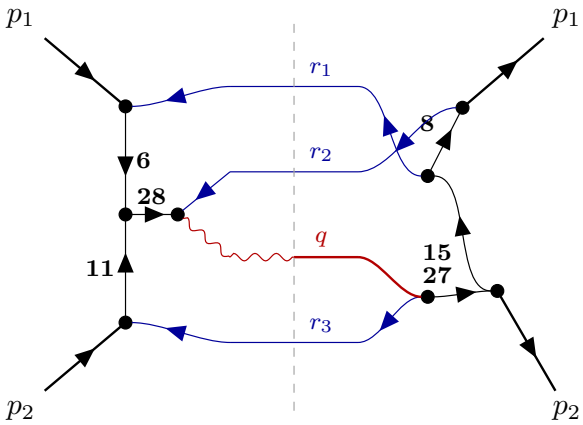
irreducible numerators: $8 = (p_1 + r_2)^2$



FAM[235] (topo235)

6	$(p_1 + r_1)^2$	amplitude
13	$(r_2 + r_3)^2$	amplitude
24	$(p_1 + p_2 + r_1)^2$	amplitude
8	$(p_1 + r_2)^2$	conjugate
11	$(p_2 + r_3)^2$	conjugate
29	$(p_1 + p_2 + r_2 + r_3)^2$	conjugate

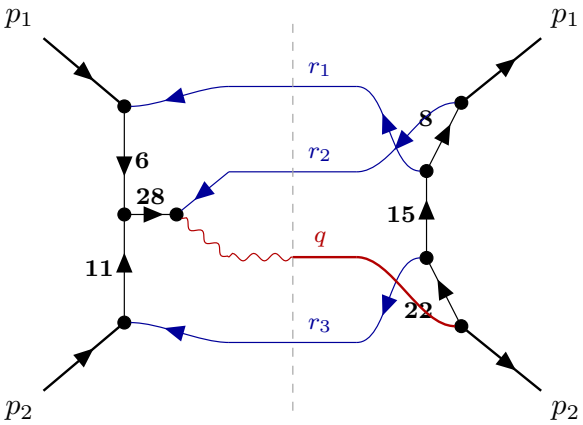
irreducible numerators: $27 = (p_1 + p_2 + r_1 + r_2)^2$



FAM[241] (topo241)

6	$(p_1 + r_1)^2$	amplitude
11	$(p_2 + r_3)^2$	amplitude
28	$(p_1 + p_2 + r_1 + r_3)^2$	amplitude
8	$(p_1 + r_2)^2$	conjugate
15	$(p_1 + r_1 + r_2)^2$	conjugate
27	$(p_1 + p_2 + r_1 + r_2)^2$	conjugate

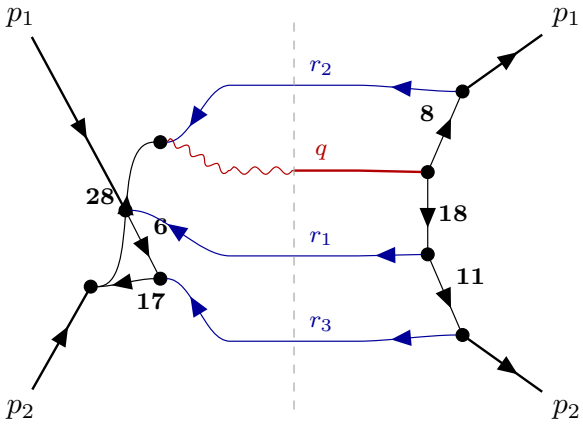
irreducible numerators: $17 = (p_1 + r_1 + r_3)^2$



FAM[247] (topo247)

6	$(p_1 + r_1)^2$	amplitude
11	$(p_2 + r_3)^2$	amplitude
28	$(p_1 + p_2 + r_1 + r_3)^2$	amplitude
8	$(p_1 + r_2)^2$	conjugate
15	$(p_1 + r_1 + r_2)^2$	conjugate
22	$(p_1 + r_1 + r_2 + r_3)^2$	conjugate

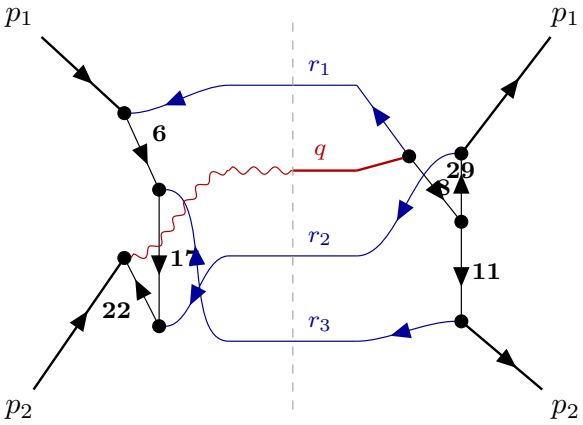
irreducible numerators: $24 = (p_1 + p_2 + r_1)^2$



FAM[253] (topo253)

6	$(p_1 + r_1)^2$	amplitude
17	$(p_1 + r_1 + r_3)^2$	amplitude
28	$(p_1 + p_2 + r_1 + r_3)^2$	amplitude
8	$(p_1 + r_2)^2$	conjugate
11	$(p_2 + r_3)^2$	conjugate
18	$(p_2 + r_1 + r_3)^2$	conjugate

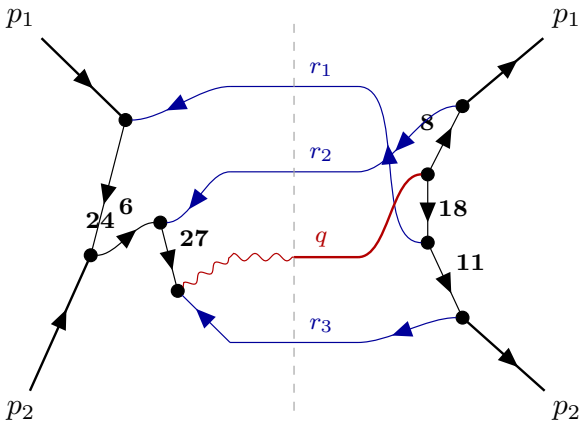
irreducible numerators: $29 = (p_1 + p_2 + r_2 + r_3)^2$



FAM[259] (topo259)

6	$(p_1 + r_1)^2$	amplitude
17	$(p_1 + r_1 + r_3)^2$	amplitude
22	$(p_1 + r_1 + r_2 + r_3)^2$	amplitude
8	$(p_1 + r_2)^2$	conjugate
11	$(p_2 + r_3)^2$	conjugate
29	$(p_1 + p_2 + r_2 + r_3)^2$	conjugate

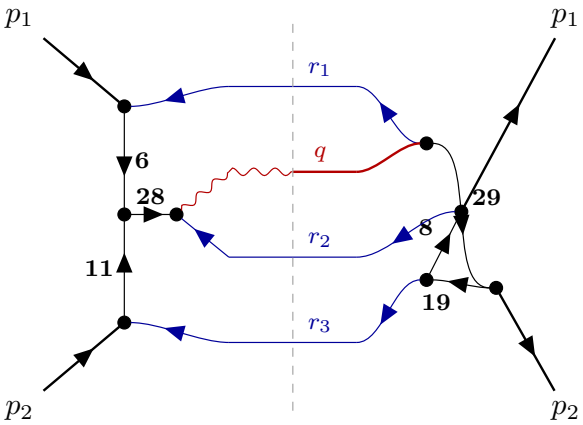
irreducible numerators: $25 = (p_1 + p_2 + r_2)^2$



FAM[265] (topo265)

6	$(p_1 + r_1)^2$	amplitude
24	$(p_1 + p_2 + r_1)^2$	amplitude
27	$(p_1 + p_2 + r_1 + r_2)^2$	amplitude
8	$(p_1 + r_2)^2$	conjugate
11	$(p_2 + r_3)^2$	conjugate
18	$(p_2 + r_1 + r_3)^2$	conjugate

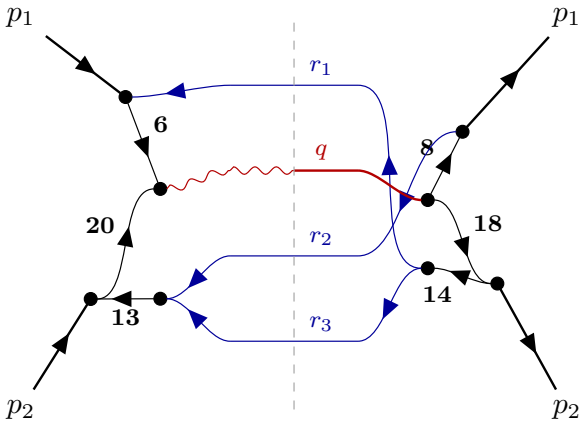
irreducible numerators: $23 = (p_2 + r_1 + r_2 + r_3)^2$



FAM[271] (topo271)

6	$(p_1 + r_1)^2$	amplitude
11	$(p_2 + r_3)^2$	amplitude
28	$(p_1 + p_2 + r_1 + r_3)^2$	amplitude
8	$(p_1 + r_2)^2$	conjugate
19	$(p_1 + r_2 + r_3)^2$	conjugate
29	$(p_1 + p_2 + r_2 + r_3)^2$	conjugate

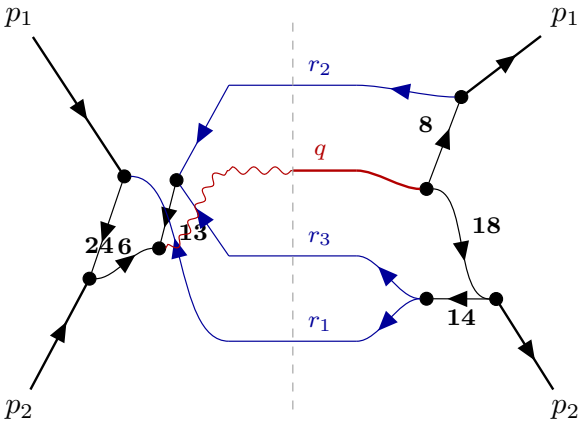
irreducible numerators: $24 = (p_1 + p_2 + r_1)^2$



FAM[277] (topo277)

6	$(p_1 + r_1)^2$	amplitude
13	$(r_2 + r_3)^2$	amplitude
20	$(p_2 + r_2 + r_3)^2$	amplitude
8	$(p_1 + r_2)^2$	conjugate
14	$(r_1 + r_3)^2$	conjugate
18	$(p_2 + r_1 + r_3)^2$	conjugate

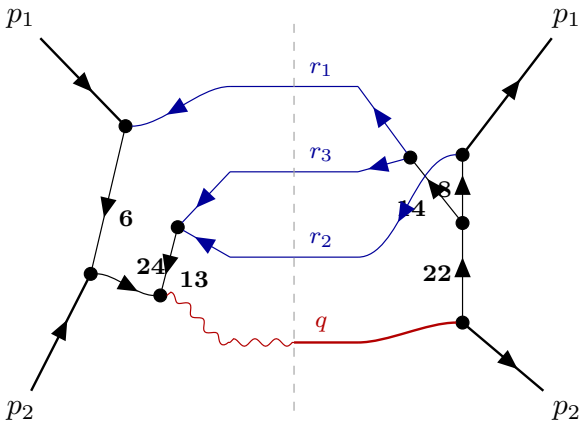
irreducible numerators: $21 = (r_1 + r_2 + r_3)^2$



FAM[280] (topo280)

6	$(p_1 + r_1)^2$	amplitude
13	$(r_2 + r_3)^2$	amplitude
24	$(p_1 + p_2 + r_1)^2$	amplitude
8	$(p_1 + r_2)^2$	conjugate
14	$(r_1 + r_3)^2$	conjugate
18	$(p_2 + r_1 + r_3)^2$	conjugate

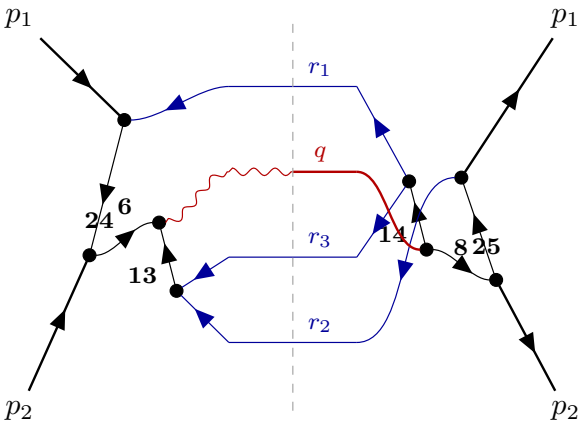
irreducible numerators: $28 = (p_1 + p_2 + r_1 + r_3)^2$



FAM[286] (topo286)

6	$(p_1 + r_1)^2$	amplitude
13	$(r_2 + r_3)^2$	amplitude
24	$(p_1 + p_2 + r_1)^2$	amplitude
8	$(p_1 + r_2)^2$	conjugate
14	$(r_1 + r_3)^2$	conjugate
22	$(p_1 + r_1 + r_2 + r_3)^2$	conjugate

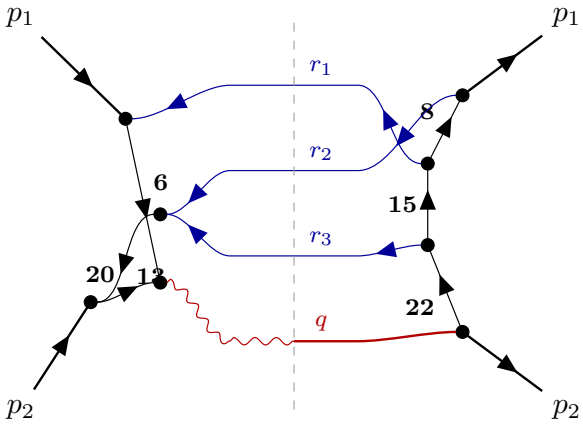
irreducible numerators: $27 = (p_1 + p_2 + r_1 + r_2)^2$



FAM[292] (topo292)

6	$(p_1 + r_1)^2$	amplitude
13	$(r_2 + r_3)^2$	amplitude
24	$(p_1 + p_2 + r_1)^2$	amplitude
8	$(p_1 + r_2)^2$	conjugate
14	$(r_1 + r_3)^2$	conjugate
25	$(p_1 + p_2 + r_2)^2$	conjugate

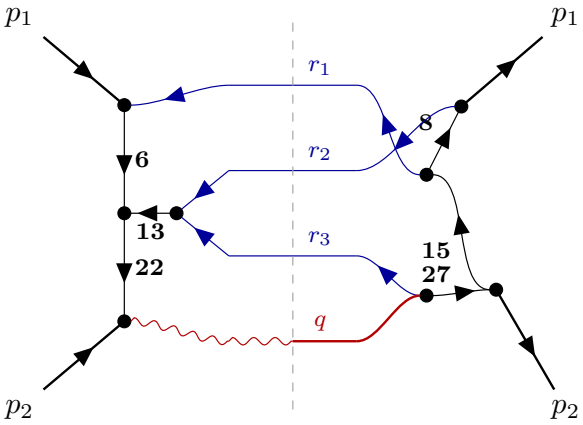
irreducible numerators: $17 = (p_1 + r_1 + r_3)^2$



FAM[298] (topo298)

6	$(p_1 + r_1)^2$	amplitude
13	$(r_2 + r_3)^2$	amplitude
20	$(p_2 + r_2 + r_3)^2$	amplitude
8	$(p_1 + r_2)^2$	conjugate
15	$(p_1 + r_1 + r_2)^2$	conjugate
22	$(p_1 + r_1 + r_2 + r_3)^2$	conjugate

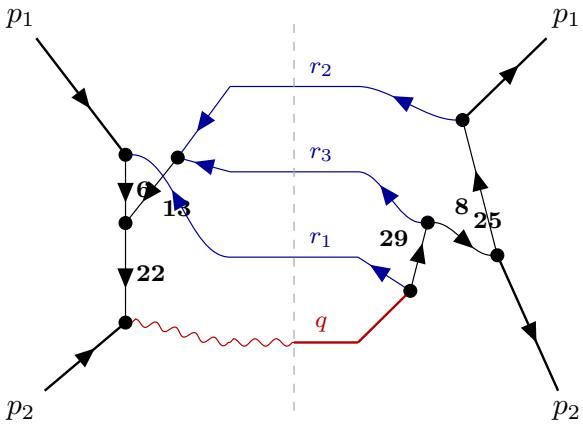
irreducible numerators: $27 = (p_1 + p_2 + r_1 + r_2)^2$



FAM[304] (topo304)

6	$(p_1 + r_1)^2$	amplitude
13	$(r_2 + r_3)^2$	amplitude
22	$(p_1 + r_1 + r_2 + r_3)^2$	amplitude
8	$(p_1 + r_2)^2$	conjugate
15	$(p_1 + r_1 + r_2)^2$	conjugate
27	$(p_1 + p_2 + r_1 + r_2)^2$	conjugate

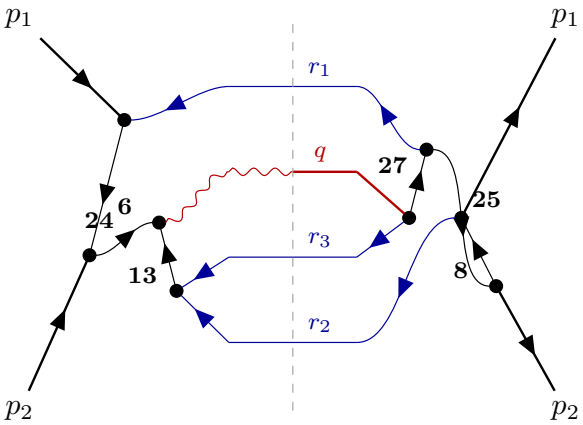
irreducible numerators: $24 = (p_1 + p_2 + r_1)^2$



FAM[310] (topo310)

6	$(p_1 + r_1)^2$	amplitude
13	$(r_2 + r_3)^2$	amplitude
22	$(p_1 + r_1 + r_2 + r_3)^2$	amplitude
8	$(p_1 + r_2)^2$	conjugate
25	$(p_1 + p_2 + r_2)^2$	conjugate
29	$(p_1 + p_2 + r_2 + r_3)^2$	conjugate

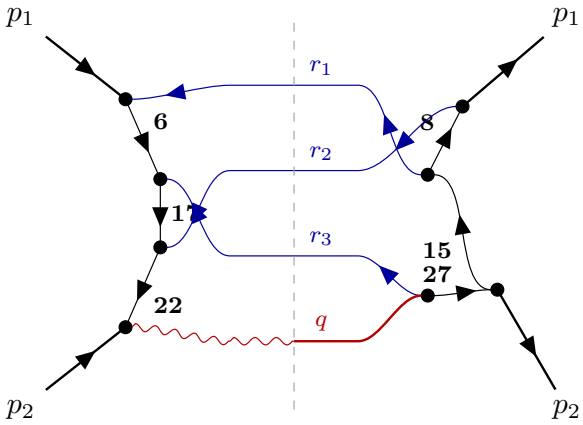
irreducible numerators: $12 = (r_1 + r_2)^2$



FAM[316] (topo316)

6	$(p_1 + r_1)^2$	amplitude
13	$(r_2 + r_3)^2$	amplitude
24	$(p_1 + p_2 + r_1)^2$	amplitude
8	$(p_1 + r_2)^2$	conjugate
25	$(p_1 + p_2 + r_2)^2$	conjugate
27	$(p_1 + p_2 + r_1 + r_2)^2$	conjugate

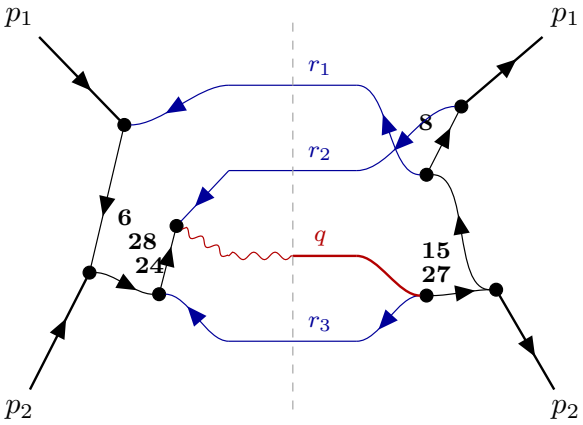
irreducible numerators: $29 = (p_1 + p_2 + r_2 + r_3)^2$



FAM[322] (topo322)

6	$(p_1 + r_1)^2$	amplitude
17	$(p_1 + r_1 + r_3)^2$	amplitude
22	$(p_1 + r_1 + r_2 + r_3)^2$	amplitude
8	$(p_1 + r_2)^2$	conjugate
15	$(p_1 + r_1 + r_2)^2$	conjugate
27	$(p_1 + p_2 + r_1 + r_2)^2$	conjugate

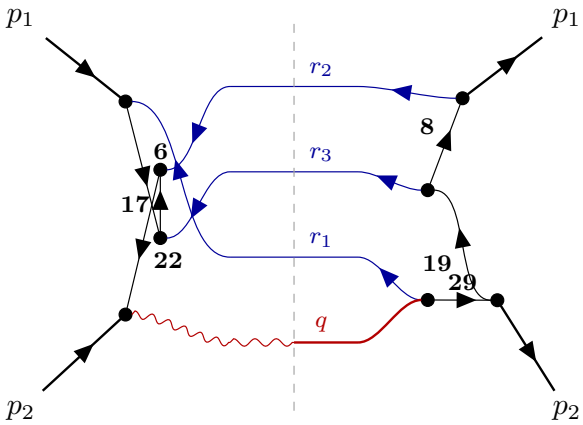
irreducible numerators: $28 = (p_1 + p_2 + r_1 + r_3)^2$



FAM[328] (topo328)

6	$(p_1 + r_1)^2$	amplitude
24	$(p_1 + p_2 + r_1)^2$	amplitude
28	$(p_1 + p_2 + r_1 + r_3)^2$	amplitude
8	$(p_1 + r_2)^2$	conjugate
15	$(p_1 + r_1 + r_2)^2$	conjugate
27	$(p_1 + p_2 + r_1 + r_2)^2$	conjugate

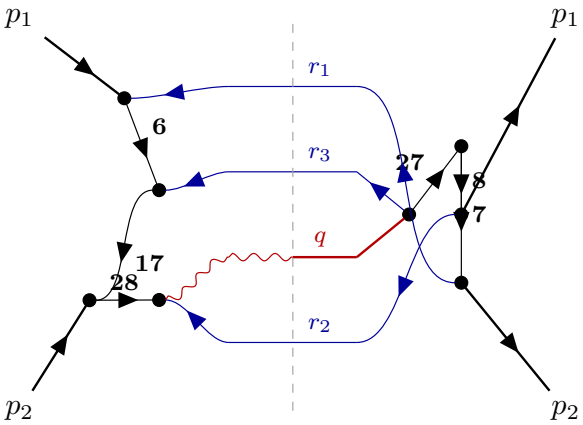
irreducible numerators: $10 = (p_1 + r_3)^2$



FAM[334] (topo334)

6	$(p_1 + r_1)^2$	amplitude
17	$(p_1 + r_1 + r_3)^2$	amplitude
22	$(p_1 + r_1 + r_2 + r_3)^2$	amplitude
8	$(p_1 + r_2)^2$	conjugate
19	$(p_1 + r_2 + r_3)^2$	conjugate
29	$(p_1 + p_2 + r_2 + r_3)^2$	conjugate

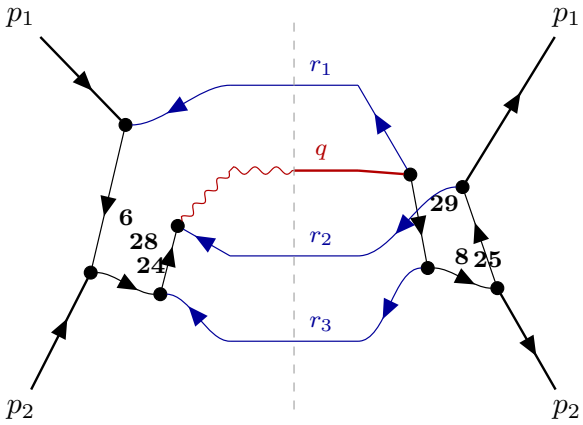
irreducible numerators: $28 = (p_1 + p_2 + r_1 + r_3)^2$



FAM[337] (topo337)

6	$(p_1 + r_1)^2$	amplitude
17	$(p_1 + r_1 + r_3)^2$	amplitude
28	$(p_1 + p_2 + r_1 + r_3)^2$	amplitude
7	$(p_2 + r_1)^2$	conjugate
8	$(p_1 + r_2)^2$	conjugate
27	$(p_1 + p_2 + r_1 + r_2)^2$	conjugate

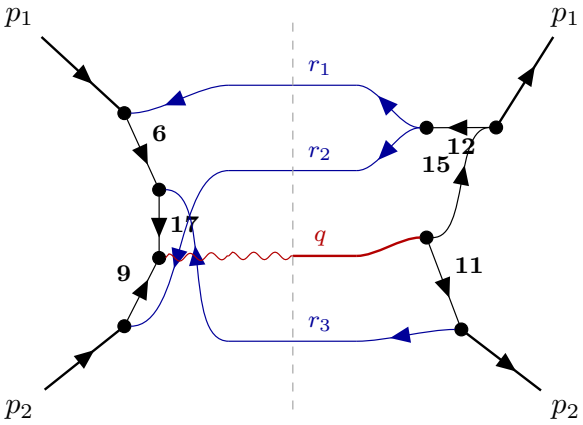
irreducible numerators: $25 = (p_1 + p_2 + r_2)^2$



FAM[343] (topo343)

6	$(p_1 + r_1)^2$	amplitude
24	$(p_1 + p_2 + r_1)^2$	amplitude
28	$(p_1 + p_2 + r_1 + r_3)^2$	amplitude
8	$(p_1 + r_2)^2$	conjugate
25	$(p_1 + p_2 + r_2)^2$	conjugate
29	$(p_1 + p_2 + r_2 + r_3)^2$	conjugate

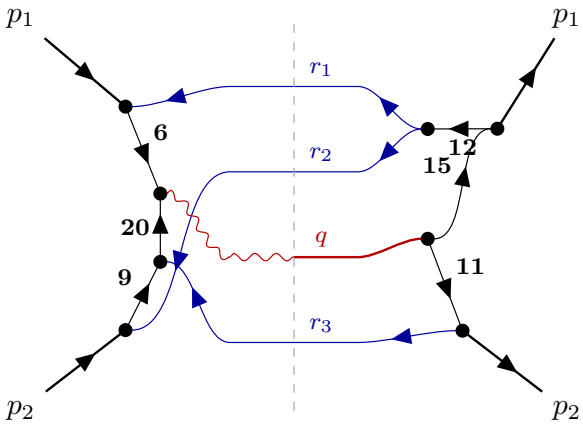
irreducible numerators: $10 = (p_1 + r_3)^2$



FAM[346] (topo346)

6	$(p_1 + r_1)^2$	amplitude
9	$(p_2 + r_2)^2$	amplitude
17	$(p_1 + r_1 + r_3)^2$	amplitude
11	$(p_2 + r_3)^2$	conjugate
12	$(r_1 + r_2)^2$	conjugate
15	$(p_1 + r_1 + r_2)^2$	conjugate

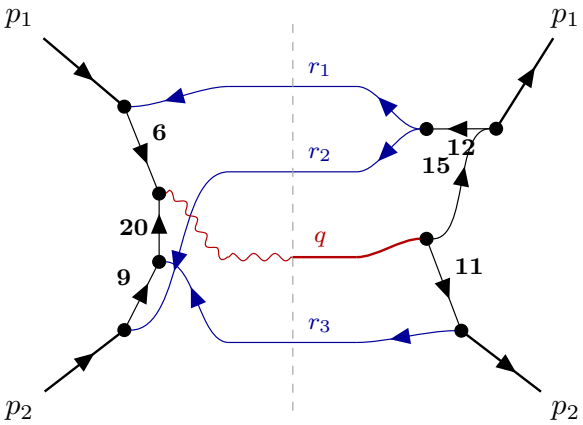
irreducible numerators: $16 = (p_2 + r_1 + r_2)^2$



FAM[352] (topo352)

6	$(p_1 + r_1)^2$	amplitude
9	$(p_2 + r_2)^2$	amplitude
20	$(p_2 + r_2 + r_3)^2$	amplitude
11	$(p_2 + r_3)^2$	conjugate
12	$(r_1 + r_2)^2$	conjugate
15	$(p_1 + r_1 + r_2)^2$	conjugate

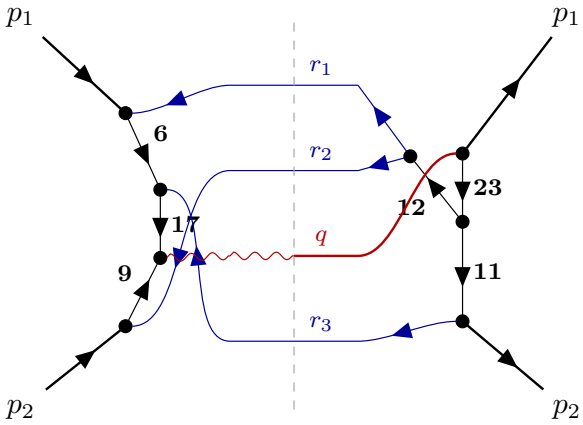
irreducible numerators: $21 = (r_1 + r_2 + r_3)^2$



FAM[358] (topo358)

6	$(p_1 + r_1)^2$	amplitude
9	$(p_2 + r_2)^2$	amplitude
20	$(p_2 + r_2 + r_3)^2$	amplitude
11	$(p_2 + r_3)^2$	conjugate
12	$(r_1 + r_2)^2$	conjugate
15	$(p_1 + r_1 + r_2)^2$	conjugate

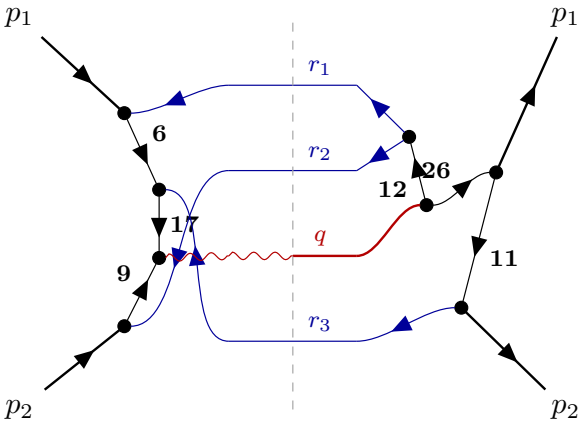
irreducible numerators: $27 = (p_1 + p_2 + r_1 + r_2)^2$



FAM[364] (topo364)

6	$(p_1 + r_1)^2$	amplitude
9	$(p_2 + r_2)^2$	amplitude
17	$(p_1 + r_1 + r_3)^2$	amplitude
11	$(p_2 + r_3)^2$	conjugate
12	$(r_1 + r_2)^2$	conjugate
23	$(p_2 + r_1 + r_2 + r_3)^2$	conjugate

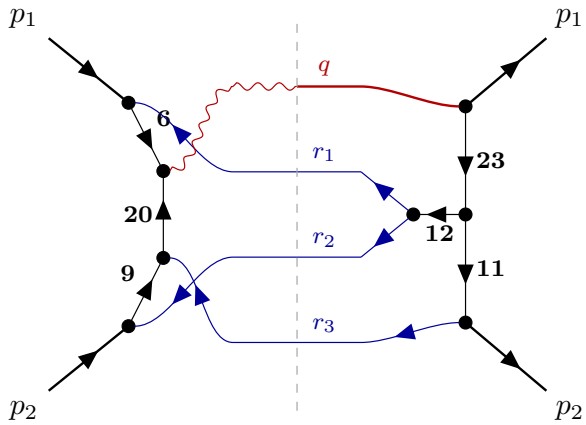
irreducible numerators: $18 = (p_2 + r_1 + r_3)^2$



FAM[370] (topo370)

6	$(p_1 + r_1)^2$	amplitude
9	$(p_2 + r_2)^2$	amplitude
17	$(p_1 + r_1 + r_3)^2$	amplitude
11	$(p_2 + r_3)^2$	conjugate
12	$(r_1 + r_2)^2$	conjugate
26	$(p_1 + p_2 + r_3)^2$	conjugate

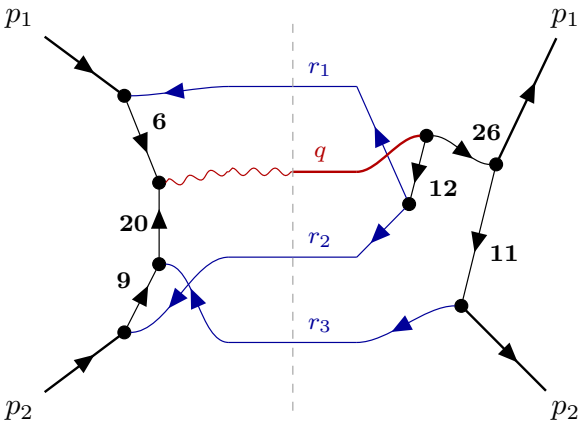
irreducible numerators: $28 = (p_1 + p_2 + r_1 + r_3)^2$



FAM[376] (topo376)

6	$(p_1 + r_1)^2$	amplitude
9	$(p_2 + r_2)^2$	amplitude
20	$(p_2 + r_2 + r_3)^2$	amplitude
11	$(p_2 + r_3)^2$	conjugate
12	$(r_1 + r_2)^2$	conjugate
23	$(p_2 + r_1 + r_2 + r_3)^2$	conjugate

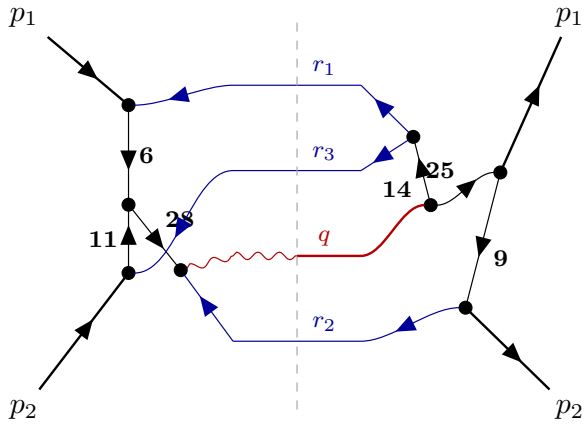
irreducible numerators: $27 = (p_1 + p_2 + r_1 + r_2)^2$



FAM[382] (topo382)

6	$(p_1 + r_1)^2$	amplitude
9	$(p_2 + r_2)^2$	amplitude
20	$(p_2 + r_2 + r_3)^2$	amplitude
11	$(p_2 + r_3)^2$	conjugate
12	$(r_1 + r_2)^2$	conjugate
26	$(p_1 + p_2 + r_3)^2$	conjugate

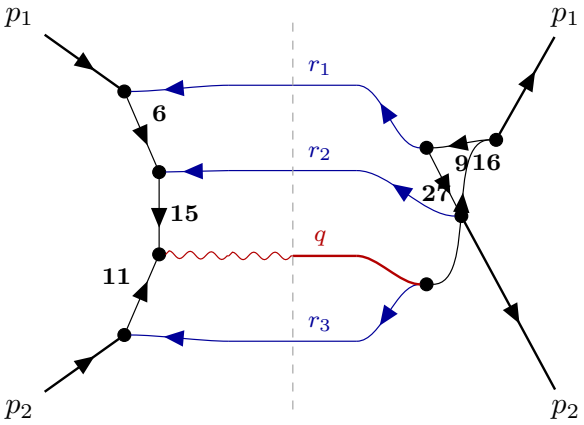
irreducible numerators: $29 = (p_1 + p_2 + r_2 + r_3)^2$



FAM[388] (topo388)

6	$(p_1 + r_1)^2$	amplitude
11	$(p_2 + r_3)^2$	amplitude
28	$(p_1 + p_2 + r_1 + r_3)^2$	amplitude
9	$(p_2 + r_2)^2$	conjugate
14	$(r_1 + r_3)^2$	conjugate
25	$(p_1 + p_2 + r_2)^2$	conjugate

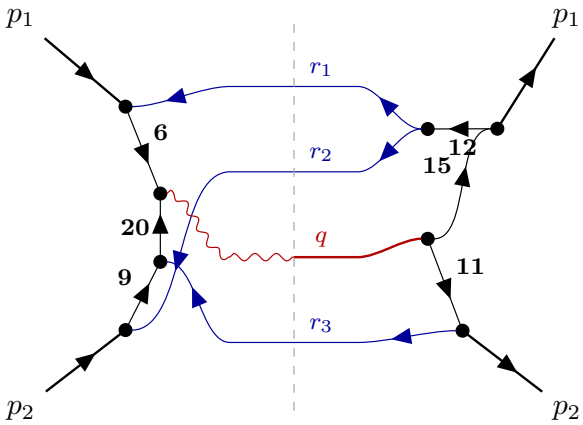
irreducible numerators: $27 = (p_1 + p_2 + r_1 + r_2)^2$



FAM[394] (topo394)

6	$(p_1 + r_1)^2$	amplitude
11	$(p_2 + r_3)^2$	amplitude
15	$(p_1 + r_1 + r_2)^2$	amplitude
9	$(p_2 + r_2)^2$	conjugate
16	$(p_2 + r_1 + r_2)^2$	conjugate
27	$(p_1 + p_2 + r_1 + r_2)^2$	conjugate

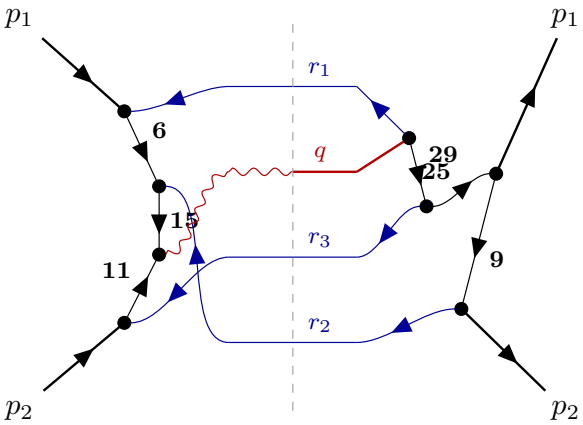
irreducible numerators: $28 = (p_1 + p_2 + r_1 + r_3)^2$



FAM[400] (topo400)

6	$(p_1 + r_1)^2$	amplitude
9	$(p_2 + r_2)^2$	amplitude
20	$(p_2 + r_2 + r_3)^2$	amplitude
11	$(p_2 + r_3)^2$	conjugate
12	$(r_1 + r_2)^2$	conjugate
15	$(p_1 + r_1 + r_2)^2$	conjugate

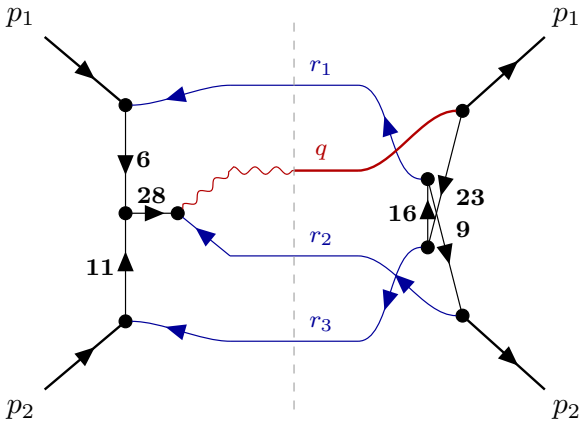
irreducible numerators: $29 = (p_1 + p_2 + r_2 + r_3)^2$



FAM[406] (topo406)

6	$(p_1 + r_1)^2$	amplitude
11	$(p_2 + r_3)^2$	amplitude
15	$(p_1 + r_1 + r_2)^2$	amplitude
9	$(p_2 + r_2)^2$	conjugate
25	$(p_1 + p_2 + r_2)^2$	conjugate
29	$(p_1 + p_2 + r_2 + r_3)^2$	conjugate

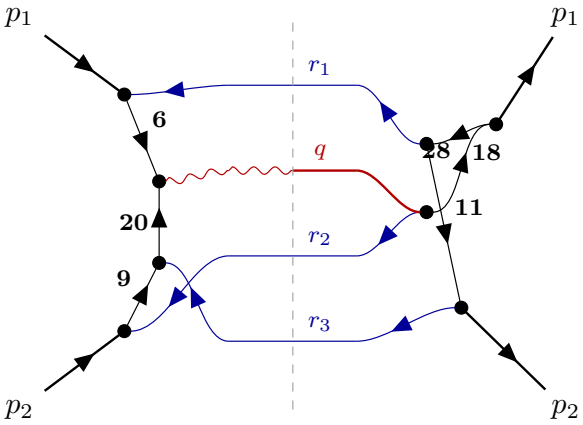
irreducible numerators: $22 = (p_1 + r_1 + r_2 + r_3)^2$



FAM[412] (topo412)

6	$(p_1 + r_1)^2$	amplitude
11	$(p_2 + r_3)^2$	amplitude
28	$(p_1 + p_2 + r_1 + r_3)^2$	amplitude
9	$(p_2 + r_2)^2$	conjugate
16	$(p_2 + r_1 + r_2)^2$	conjugate
23	$(p_2 + r_1 + r_2 + r_3)^2$	conjugate

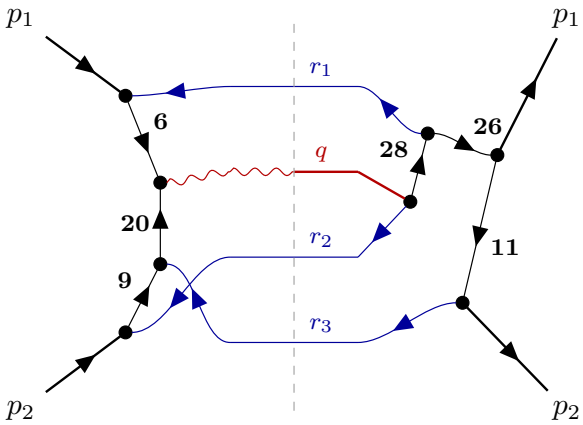
irreducible numerators: $26 = (p_1 + p_2 + r_3)^2$



FAM[418] (topo418)

6	$(p_1 + r_1)^2$	amplitude
9	$(p_2 + r_2)^2$	amplitude
20	$(p_2 + r_2 + r_3)^2$	amplitude
11	$(p_2 + r_3)^2$	conjugate
18	$(p_2 + r_1 + r_3)^2$	conjugate
28	$(p_1 + p_2 + r_1 + r_3)^2$	conjugate

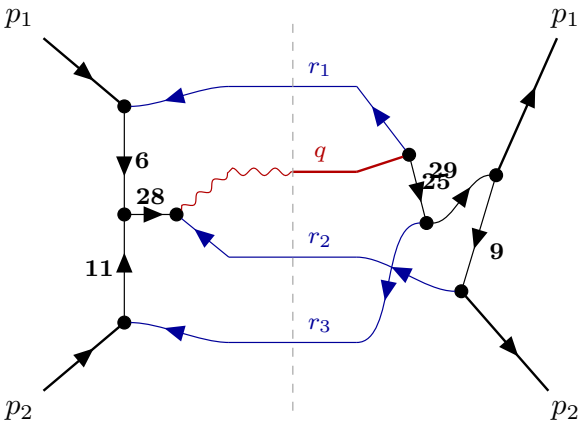
irreducible numerators: $14 = (r_1 + r_3)^2$



FAM[424] (topo424)

6	$(p_1 + r_1)^2$	amplitude
9	$(p_2 + r_2)^2$	amplitude
20	$(p_2 + r_2 + r_3)^2$	amplitude
11	$(p_2 + r_3)^2$	conjugate
26	$(p_1 + p_2 + r_3)^2$	conjugate
28	$(p_1 + p_2 + r_1 + r_3)^2$	conjugate

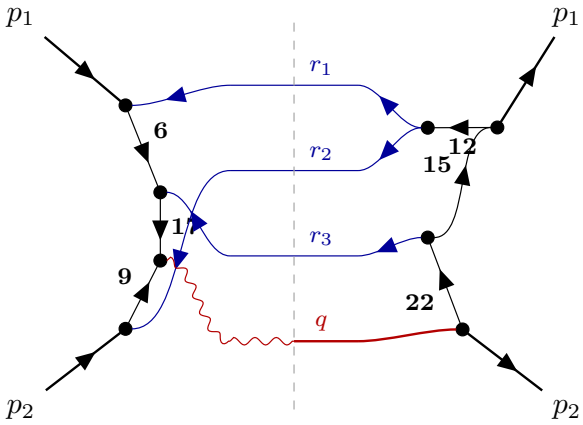
irreducible numerators: $23 = (p_2 + r_1 + r_2 + r_3)^2$



FAM[430] (topo430)

6	$(p_1 + r_1)^2$	amplitude
11	$(p_2 + r_3)^2$	amplitude
28	$(p_1 + p_2 + r_1 + r_3)^2$	amplitude
9	$(p_2 + r_2)^2$	conjugate
25	$(p_1 + p_2 + r_2)^2$	conjugate
29	$(p_1 + p_2 + r_2 + r_3)^2$	conjugate

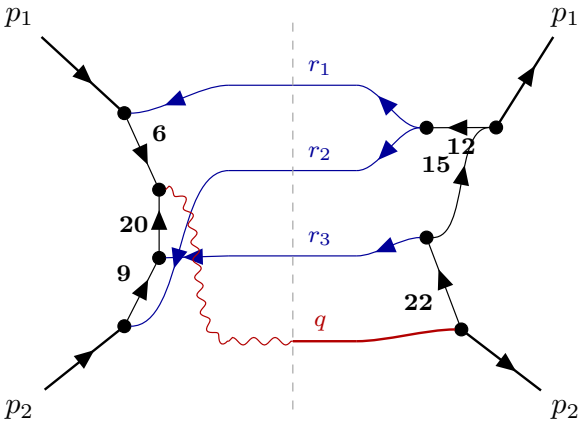
irreducible numerators: $17 = (p_1 + r_1 + r_3)^2$



FAM[436] (topo436)

6	$(p_1 + r_1)^2$	amplitude
9	$(p_2 + r_2)^2$	amplitude
17	$(p_1 + r_1 + r_3)^2$	amplitude
12	$(r_1 + r_2)^2$	conjugate
15	$(p_1 + r_1 + r_2)^2$	conjugate
22	$(p_1 + r_1 + r_2 + r_3)^2$	conjugate

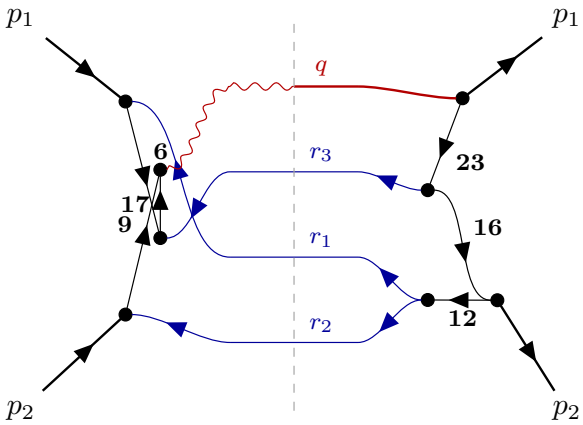
irreducible numerators: $27 = (p_1 + p_2 + r_1 + r_2)^2$



FAM[442] (topo442)

6	$(p_1 + r_1)^2$	amplitude
9	$(p_2 + r_2)^2$	amplitude
20	$(p_2 + r_2 + r_3)^2$	amplitude
12	$(r_1 + r_2)^2$	conjugate
15	$(p_1 + r_1 + r_2)^2$	conjugate
22	$(p_1 + r_1 + r_2 + r_3)^2$	conjugate

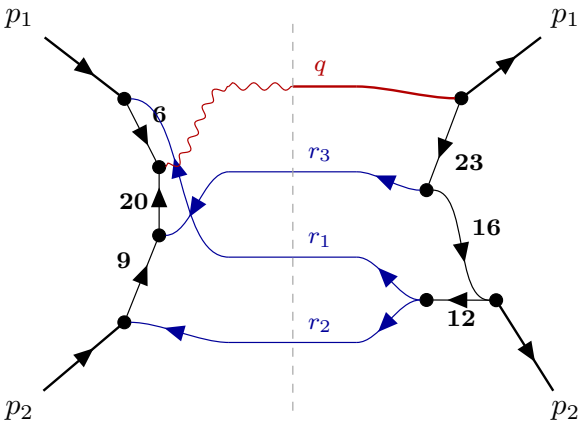
irreducible numerators: $16 = (p_2 + r_1 + r_2)^2$



FAM[448] (topo448)

6	$(p_1 + r_1)^2$	amplitude
9	$(p_2 + r_2)^2$	amplitude
17	$(p_1 + r_1 + r_3)^2$	amplitude
12	$(r_1 + r_2)^2$	conjugate
16	$(p_2 + r_1 + r_2)^2$	conjugate
23	$(p_2 + r_1 + r_2 + r_3)^2$	conjugate

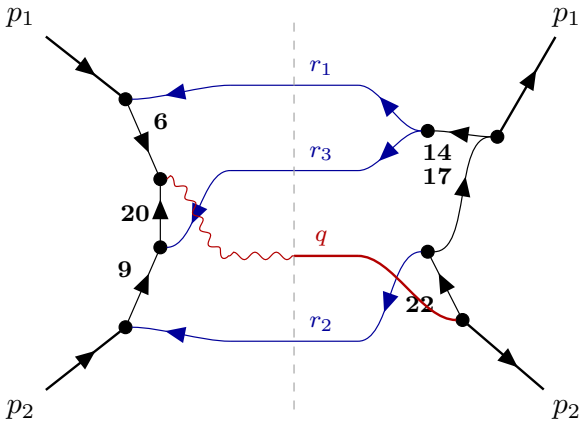
irreducible numerators: $15 = (p_1 + r_1 + r_2)^2$



FAM[454] (topo454)

6	$(p_1 + r_1)^2$	amplitude
9	$(p_2 + r_2)^2$	amplitude
20	$(p_2 + r_2 + r_3)^2$	amplitude
12	$(r_1 + r_2)^2$	conjugate
16	$(p_2 + r_1 + r_2)^2$	conjugate
23	$(p_2 + r_1 + r_2 + r_3)^2$	conjugate

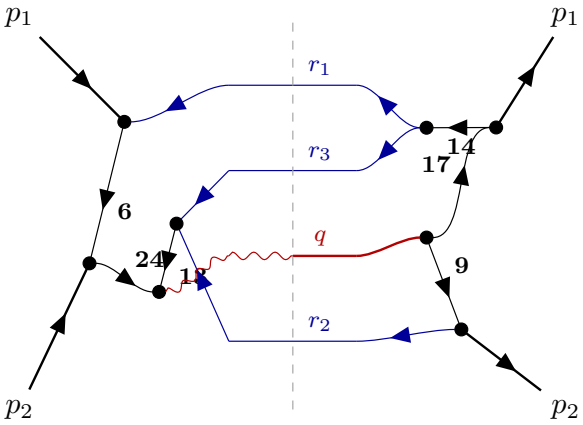
irreducible numerators: $27 = (p_1 + p_2 + r_1 + r_2)^2$



FAM[460] (topo460)

6	$(p_1 + r_1)^2$	amplitude
9	$(p_2 + r_2)^2$	amplitude
20	$(p_2 + r_2 + r_3)^2$	amplitude
14	$(r_1 + r_3)^2$	conjugate
17	$(p_1 + r_1 + r_3)^2$	conjugate
22	$(p_1 + r_1 + r_2 + r_3)^2$	conjugate

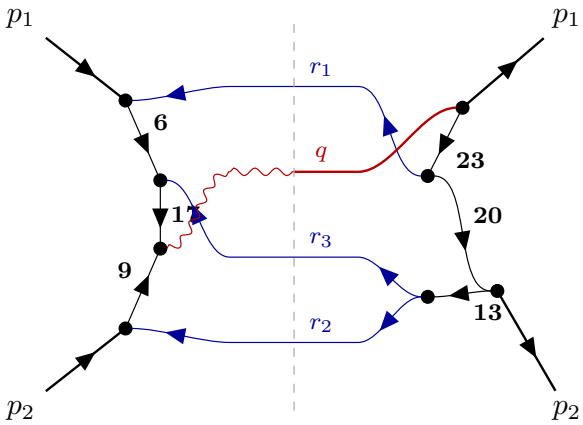
irreducible numerators: $13 = (r_2 + r_3)^2$



FAM[466] (topo466)

6	$(p_1 + r_1)^2$	amplitude
13	$(r_2 + r_3)^2$	amplitude
24	$(p_1 + p_2 + r_1)^2$	amplitude
9	$(p_2 + r_2)^2$	conjugate
14	$(r_1 + r_3)^2$	conjugate
17	$(p_1 + r_1 + r_3)^2$	conjugate

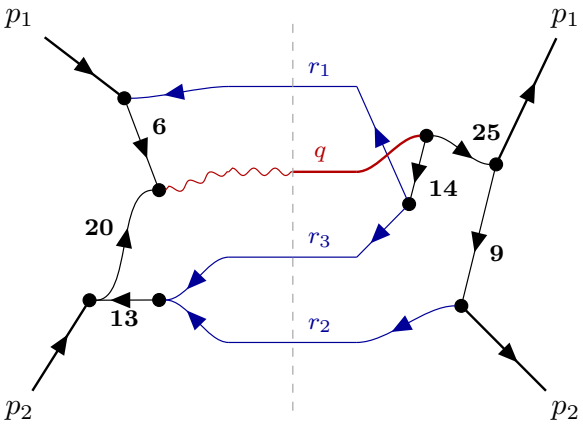
irreducible numerators: $21 = (r_1 + r_2 + r_3)^2$



FAM[472] (topo472)

6	$(p_1 + r_1)^2$	amplitude
9	$(p_2 + r_2)^2$	amplitude
17	$(p_1 + r_1 + r_3)^2$	amplitude
13	$(r_2 + r_3)^2$	conjugate
20	$(p_2 + r_2 + r_3)^2$	conjugate
23	$(p_2 + r_1 + r_2 + r_3)^2$	conjugate

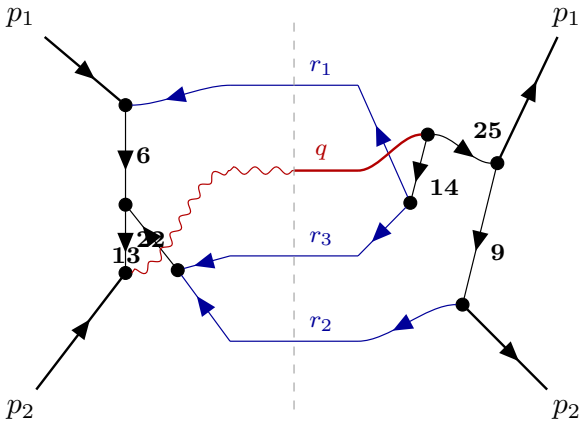
irreducible numerators: $14 = (r_1 + r_3)^2$



FAM[478] (topo478)

6	$(p_1 + r_1)^2$	amplitude
13	$(r_2 + r_3)^2$	amplitude
20	$(p_2 + r_2 + r_3)^2$	amplitude
9	$(p_2 + r_2)^2$	conjugate
14	$(r_1 + r_3)^2$	conjugate
25	$(p_1 + p_2 + r_2)^2$	conjugate

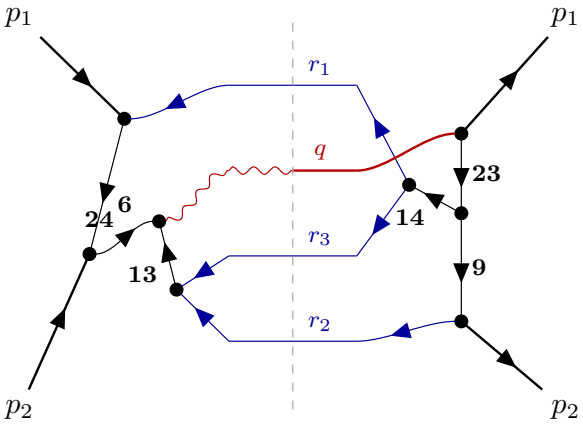
irreducible numerators: $29 = (p_1 + p_2 + r_2 + r_3)^2$



FAM[484] (topo484)

6	$(p_1 + r_1)^2$	amplitude
13	$(r_2 + r_3)^2$	amplitude
22	$(p_1 + r_1 + r_2 + r_3)^2$	amplitude
9	$(p_2 + r_2)^2$	conjugate
14	$(r_1 + r_3)^2$	conjugate
25	$(p_1 + p_2 + r_2)^2$	conjugate

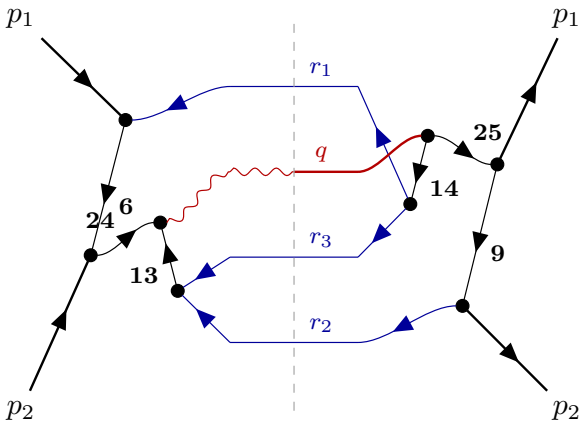
irreducible numerators: $27 = (p_1 + p_2 + r_1 + r_2)^2$



FAM[490] (topo490)

6	$(p_1 + r_1)^2$	amplitude
13	$(r_2 + r_3)^2$	amplitude
24	$(p_1 + p_2 + r_1)^2$	amplitude
9	$(p_2 + r_2)^2$	conjugate
14	$(r_1 + r_3)^2$	conjugate
23	$(p_2 + r_1 + r_2 + r_3)^2$	conjugate

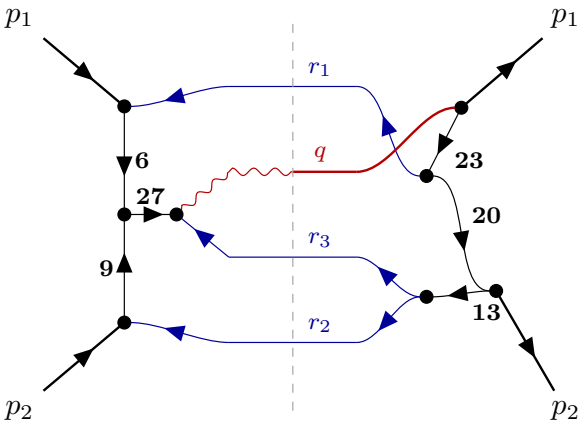
irreducible numerators: $27 = (p_1 + p_2 + r_1 + r_2)^2$



FAM[496] (topo496)

6	$(p_1 + r_1)^2$	amplitude
13	$(r_2 + r_3)^2$	amplitude
24	$(p_1 + p_2 + r_1)^2$	amplitude
9	$(p_2 + r_2)^2$	conjugate
14	$(r_1 + r_3)^2$	conjugate
25	$(p_1 + p_2 + r_2)^2$	conjugate

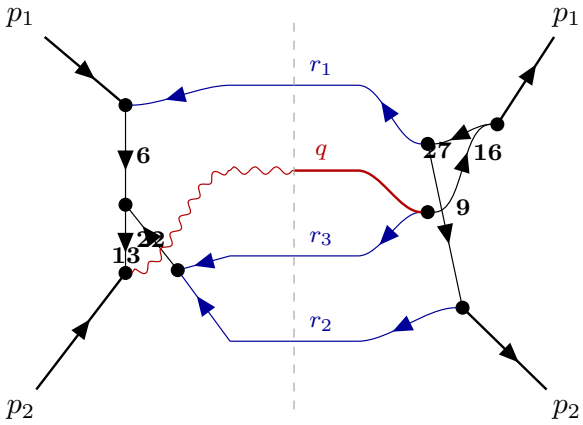
irreducible numerators: $21 = (r_1 + r_2 + r_3)^2$



FAM[502] (topo502)

6	$(p_1 + r_1)^2$	amplitude
9	$(p_2 + r_2)^2$	amplitude
27	$(p_1 + p_2 + r_1 + r_2)^2$	amplitude
13	$(r_2 + r_3)^2$	conjugate
20	$(p_2 + r_2 + r_3)^2$	conjugate
23	$(p_2 + r_1 + r_2 + r_3)^2$	conjugate

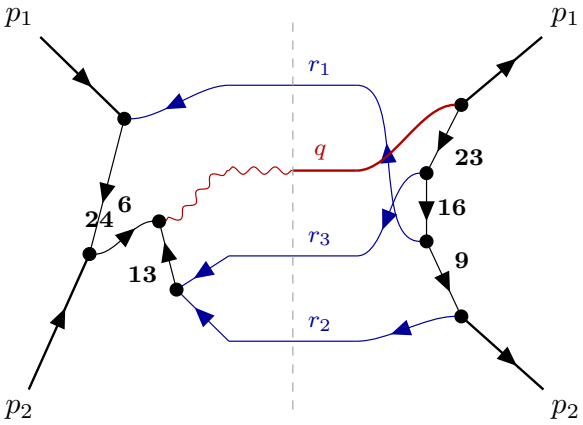
irreducible numerators: $16 = (p_2 + r_1 + r_2)^2$



FAM[508] (topo508)

6	$(p_1 + r_1)^2$	amplitude
13	$(r_2 + r_3)^2$	amplitude
22	$(p_1 + r_1 + r_2 + r_3)^2$	amplitude
9	$(p_2 + r_2)^2$	conjugate
16	$(p_2 + r_1 + r_2)^2$	conjugate
27	$(p_1 + p_2 + r_1 + r_2)^2$	conjugate

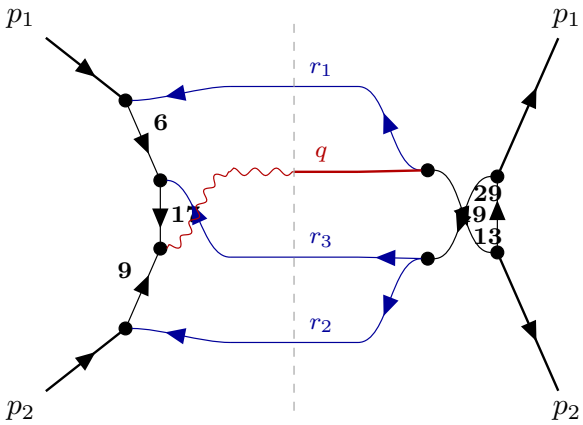
irreducible numerators: $15 = (p_1 + r_1 + r_2)^2$



FAM[514] (topo514)

6	$(p_1 + r_1)^2$	amplitude
13	$(r_2 + r_3)^2$	amplitude
24	$(p_1 + p_2 + r_1)^2$	amplitude
9	$(p_2 + r_2)^2$	conjugate
16	$(p_2 + r_1 + r_2)^2$	conjugate
23	$(p_2 + r_1 + r_2 + r_3)^2$	conjugate

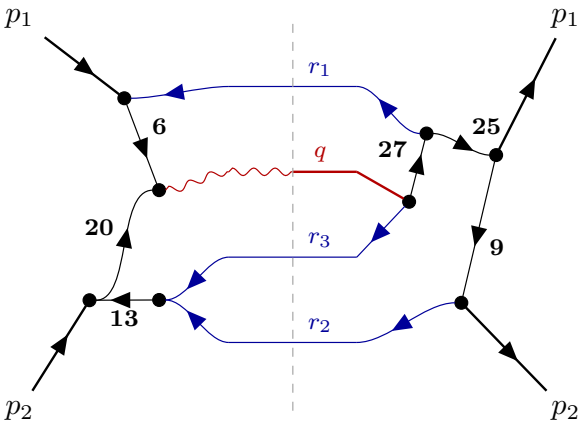
irreducible numerators: $27 = (p_1 + p_2 + r_1 + r_2)^2$



FAM[520] (topo520)

6	$(p_1 + r_1)^2$	amplitude
9	$(p_2 + r_2)^2$	amplitude
17	$(p_1 + r_1 + r_3)^2$	amplitude
13	$(r_2 + r_3)^2$	conjugate
19	$(p_1 + r_2 + r_3)^2$	conjugate
29	$(p_1 + p_2 + r_2 + r_3)^2$	conjugate

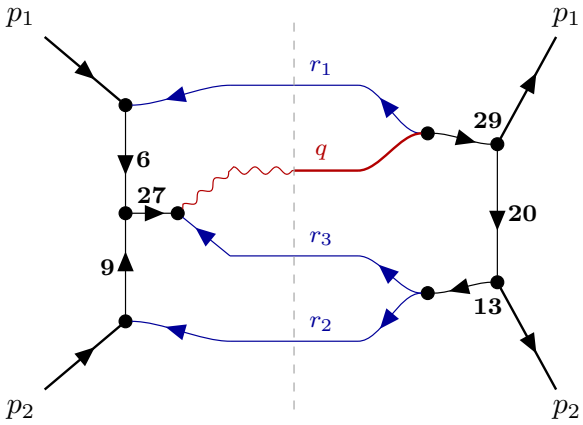
irreducible numerators: $14 = (r_1 + r_3)^2$



FAM[526] (topo526)

6	$(p_1 + r_1)^2$	amplitude
13	$(r_2 + r_3)^2$	amplitude
20	$(p_2 + r_2 + r_3)^2$	amplitude
9	$(p_2 + r_2)^2$	conjugate
25	$(p_1 + p_2 + r_2)^2$	conjugate
27	$(p_1 + p_2 + r_1 + r_2)^2$	conjugate

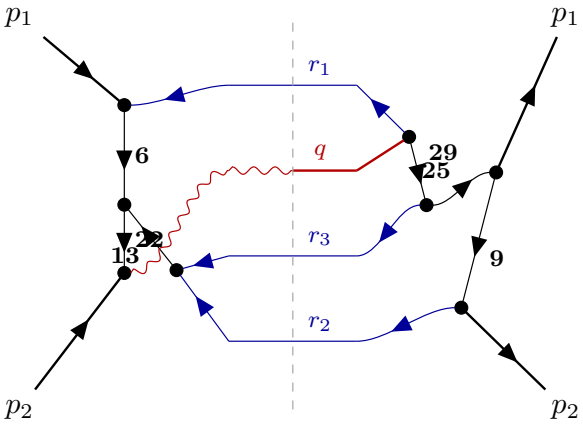
irreducible numerators: $21 = (r_1 + r_2 + r_3)^2$



FAM[532] (topo532)

6	$(p_1 + r_1)^2$	amplitude
9	$(p_2 + r_2)^2$	amplitude
27	$(p_1 + p_2 + r_1 + r_2)^2$	amplitude
13	$(r_2 + r_3)^2$	conjugate
20	$(p_2 + r_2 + r_3)^2$	conjugate
29	$(p_1 + p_2 + r_2 + r_3)^2$	conjugate

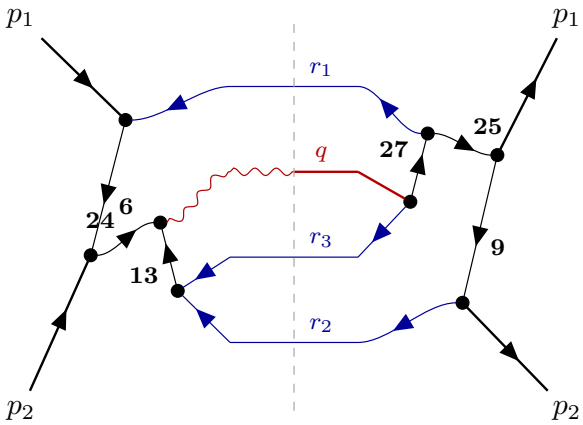
irreducible numerators: $16 = (p_2 + r_1 + r_2)^2$



FAM[538] (topo538)

6	$(p_1 + r_1)^2$	amplitude
13	$(r_2 + r_3)^2$	amplitude
22	$(p_1 + r_1 + r_2 + r_3)^2$	amplitude
9	$(p_2 + r_2)^2$	conjugate
25	$(p_1 + p_2 + r_2)^2$	conjugate
29	$(p_1 + p_2 + r_2 + r_3)^2$	conjugate

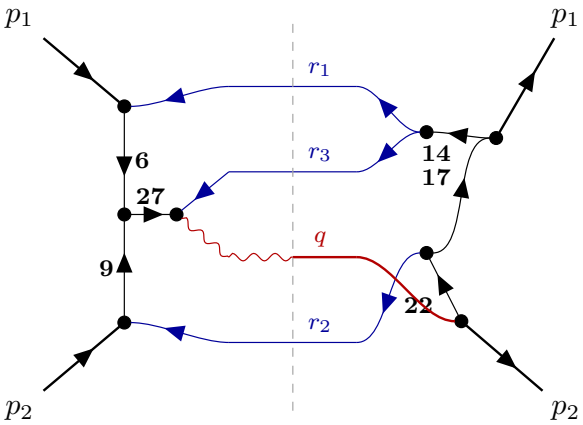
irreducible numerators: $17 = (p_1 + r_1 + r_3)^2$



FAM[544] (topo544)

6	$(p_1 + r_1)^2$	amplitude
13	$(r_2 + r_3)^2$	amplitude
24	$(p_1 + p_2 + r_1)^2$	amplitude
9	$(p_2 + r_2)^2$	conjugate
25	$(p_1 + p_2 + r_2)^2$	conjugate
27	$(p_1 + p_2 + r_1 + r_2)^2$	conjugate

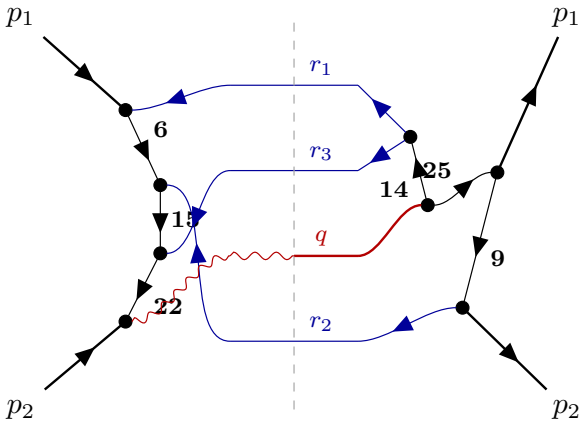
irreducible numerators: $29 = (p_1 + p_2 + r_2 + r_3)^2$



FAM[550] (topo550)

6	$(p_1 + r_1)^2$	amplitude
9	$(p_2 + r_2)^2$	amplitude
27	$(p_1 + p_2 + r_1 + r_2)^2$	amplitude
14	$(r_1 + r_3)^2$	conjugate
17	$(p_1 + r_1 + r_3)^2$	conjugate
22	$(p_1 + r_1 + r_2 + r_3)^2$	conjugate

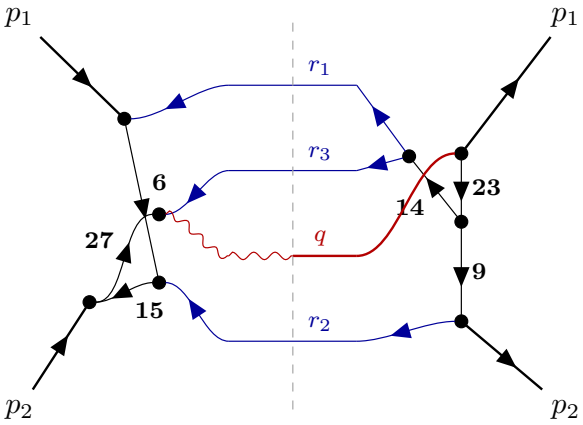
irreducible numerators: $15 = (p_1 + r_1 + r_2)^2$



FAM[556] (topo556)

6	$(p_1 + r_1)^2$	amplitude
15	$(p_1 + r_1 + r_2)^2$	amplitude
22	$(p_1 + r_1 + r_2 + r_3)^2$	amplitude
9	$(p_2 + r_2)^2$	conjugate
14	$(r_1 + r_3)^2$	conjugate
25	$(p_1 + p_2 + r_2)^2$	conjugate

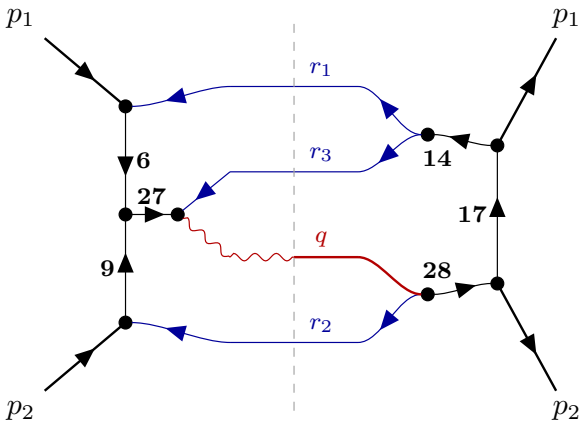
irreducible numerators: $27 = (p_1 + p_2 + r_1 + r_2)^2$



FAM[562] (topo562)

6	$(p_1 + r_1)^2$	amplitude
15	$(p_1 + r_1 + r_2)^2$	amplitude
27	$(p_1 + p_2 + r_1 + r_2)^2$	amplitude
9	$(p_2 + r_2)^2$	conjugate
14	$(r_1 + r_3)^2$	conjugate
23	$(p_2 + r_1 + r_2 + r_3)^2$	conjugate

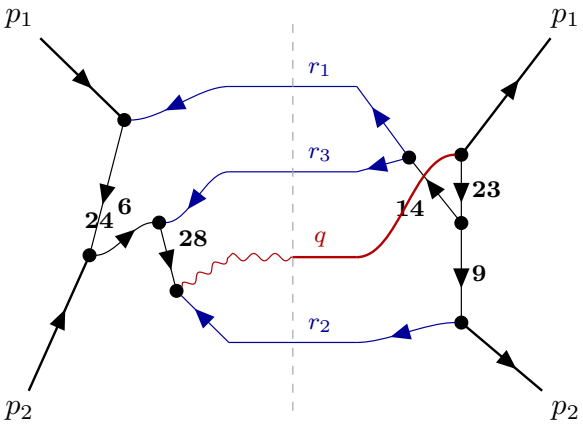
irreducible numerators: $20 = (p_2 + r_2 + r_3)^2$



FAM[568] (topo568)

6	$(p_1 + r_1)^2$	amplitude
9	$(p_2 + r_2)^2$	amplitude
27	$(p_1 + p_2 + r_1 + r_2)^2$	amplitude
14	$(r_1 + r_3)^2$	conjugate
17	$(p_1 + r_1 + r_3)^2$	conjugate
28	$(p_1 + p_2 + r_1 + r_3)^2$	conjugate

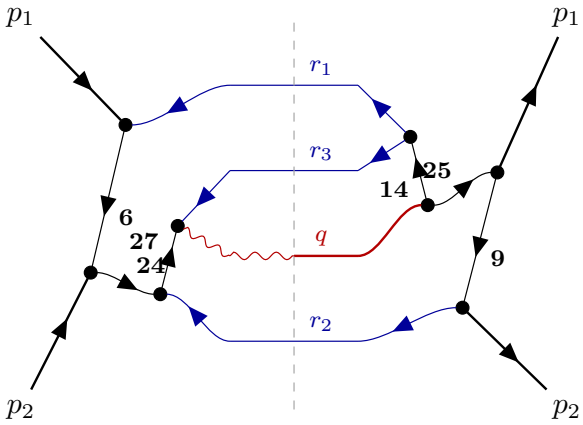
irreducible numerators: $24 = (p_1 + p_2 + r_1)^2$



FAM[574] (topo574)

6	$(p_1 + r_1)^2$	amplitude
24	$(p_1 + p_2 + r_1)^2$	amplitude
28	$(p_1 + p_2 + r_1 + r_3)^2$	amplitude
9	$(p_2 + r_2)^2$	conjugate
14	$(r_1 + r_3)^2$	conjugate
23	$(p_2 + r_1 + r_2 + r_3)^2$	conjugate

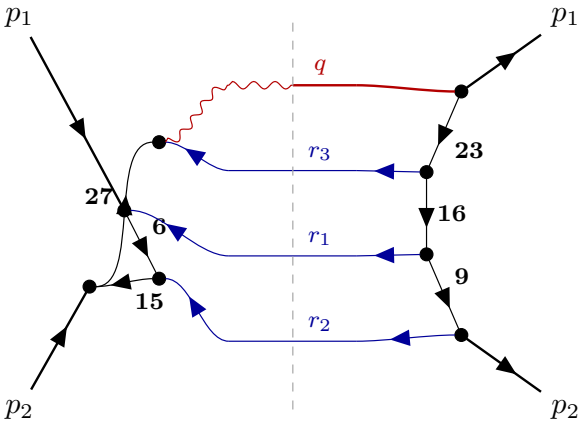
irreducible numerators: $20 = (p_2 + r_2 + r_3)^2$



FAM[580] (topo580)

6	$(p_1 + r_1)^2$	amplitude
24	$(p_1 + p_2 + r_1)^2$	amplitude
27	$(p_1 + p_2 + r_1 + r_2)^2$	amplitude
9	$(p_2 + r_2)^2$	conjugate
14	$(r_1 + r_3)^2$	conjugate
25	$(p_1 + p_2 + r_2)^2$	conjugate

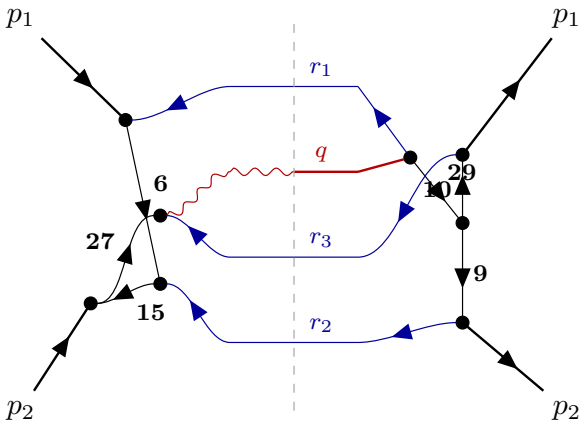
irreducible numerators: $28 = (p_1 + p_2 + r_1 + r_3)^2$



FAM[586] (topo586)

6	$(p_1 + r_1)^2$	amplitude
15	$(p_1 + r_1 + r_2)^2$	amplitude
27	$(p_1 + p_2 + r_1 + r_2)^2$	amplitude
9	$(p_2 + r_2)^2$	conjugate
16	$(p_2 + r_1 + r_2)^2$	conjugate
23	$(p_2 + r_1 + r_2 + r_3)^2$	conjugate

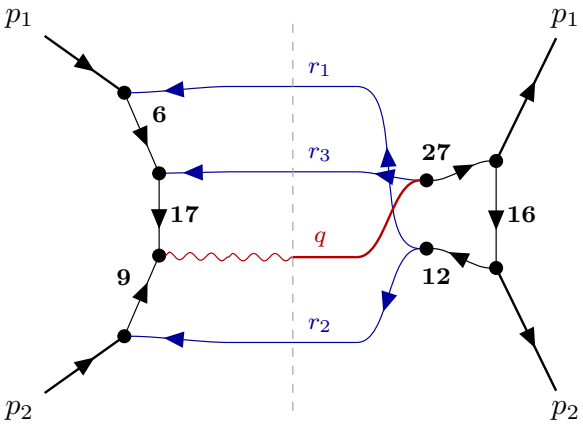
irreducible numerators: $13 = (r_2 + r_3)^2$



FAM[592] (topo592)

6	$(p_1 + r_1)^2$	amplitude
15	$(p_1 + r_1 + r_2)^2$	amplitude
27	$(p_1 + p_2 + r_1 + r_2)^2$	amplitude
9	$(p_2 + r_2)^2$	conjugate
10	$(p_1 + r_3)^2$	conjugate
29	$(p_1 + p_2 + r_2 + r_3)^2$	conjugate

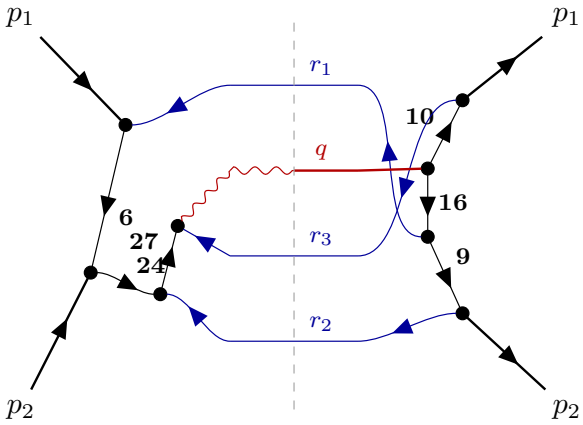
irreducible numerators: $25 = (p_1 + p_2 + r_2)^2$



FAM[598] (topo598)

6	$(p_1 + r_1)^2$	amplitude
9	$(p_2 + r_2)^2$	amplitude
17	$(p_1 + r_1 + r_3)^2$	amplitude
12	$(r_1 + r_2)^2$	conjugate
16	$(p_2 + r_1 + r_2)^2$	conjugate
27	$(p_1 + p_2 + r_1 + r_2)^2$	conjugate

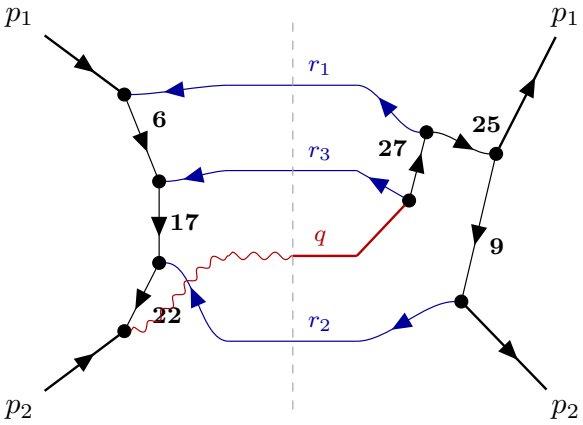
irreducible numerators: $28 = (p_1 + p_2 + r_1 + r_3)^2$



FAM[604] (topo604)

6	$(p_1 + r_1)^2$	amplitude
24	$(p_1 + p_2 + r_1)^2$	amplitude
27	$(p_1 + p_2 + r_1 + r_2)^2$	amplitude
9	$(p_2 + r_2)^2$	conjugate
10	$(p_1 + r_3)^2$	conjugate
16	$(p_2 + r_1 + r_2)^2$	conjugate

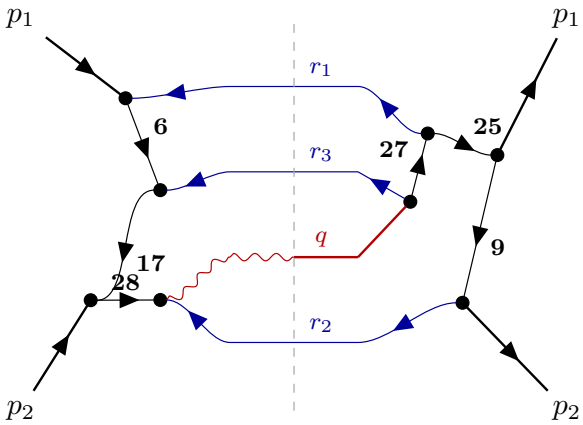
irreducible numerators: $28 = (p_1 + p_2 + r_1 + r_3)^2$



FAM[610] (topo610)

6	$(p_1 + r_1)^2$	amplitude
17	$(p_1 + r_1 + r_3)^2$	amplitude
22	$(p_1 + r_1 + r_2 + r_3)^2$	amplitude
9	$(p_2 + r_2)^2$	conjugate
25	$(p_1 + p_2 + r_2)^2$	conjugate
27	$(p_1 + p_2 + r_1 + r_2)^2$	conjugate

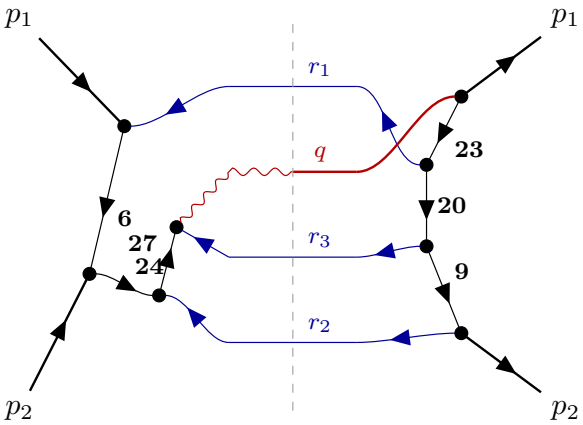
irreducible numerators: $7 = (p_2 + r_1)^2$



FAM[616] (topo616)

6	$(p_1 + r_1)^2$	amplitude
17	$(p_1 + r_1 + r_3)^2$	amplitude
28	$(p_1 + p_2 + r_1 + r_3)^2$	amplitude
9	$(p_2 + r_2)^2$	conjugate
25	$(p_1 + p_2 + r_2)^2$	conjugate
27	$(p_1 + p_2 + r_1 + r_2)^2$	conjugate

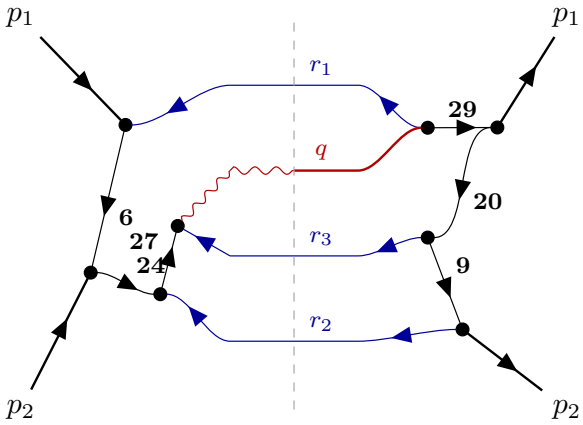
irreducible numerators: $7 = (p_2 + r_1)^2$



FAM[622] (topo622)

6	$(p_1 + r_1)^2$	amplitude
24	$(p_1 + p_2 + r_1)^2$	amplitude
27	$(p_1 + p_2 + r_1 + r_2)^2$	amplitude
9	$(p_2 + r_2)^2$	conjugate
20	$(p_2 + r_2 + r_3)^2$	conjugate
23	$(p_2 + r_1 + r_2 + r_3)^2$	conjugate

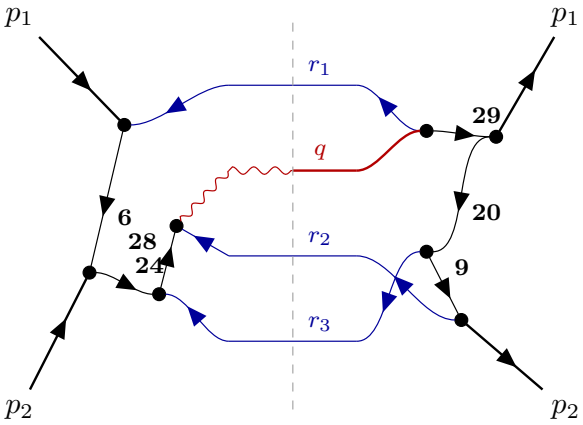
irreducible numerators: $8 = (p_1 + r_2)^2$



FAM[628] (topo628)

6	$(p_1 + r_1)^2$	amplitude
24	$(p_1 + p_2 + r_1)^2$	amplitude
27	$(p_1 + p_2 + r_1 + r_2)^2$	amplitude
9	$(p_2 + r_2)^2$	conjugate
20	$(p_2 + r_2 + r_3)^2$	conjugate
29	$(p_1 + p_2 + r_2 + r_3)^2$	conjugate

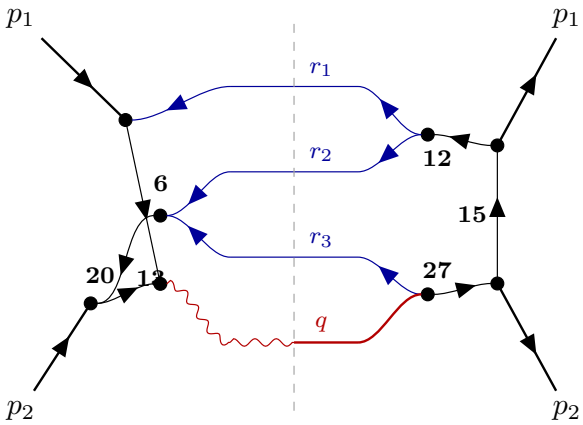
irreducible numerators: $8 = (p_1 + r_2)^2$



FAM[634] (topo634)

6	$(p_1 + r_1)^2$	amplitude
24	$(p_1 + p_2 + r_1)^2$	amplitude
28	$(p_1 + p_2 + r_1 + r_3)^2$	amplitude
9	$(p_2 + r_2)^2$	conjugate
20	$(p_2 + r_2 + r_3)^2$	conjugate
29	$(p_1 + p_2 + r_2 + r_3)^2$	conjugate

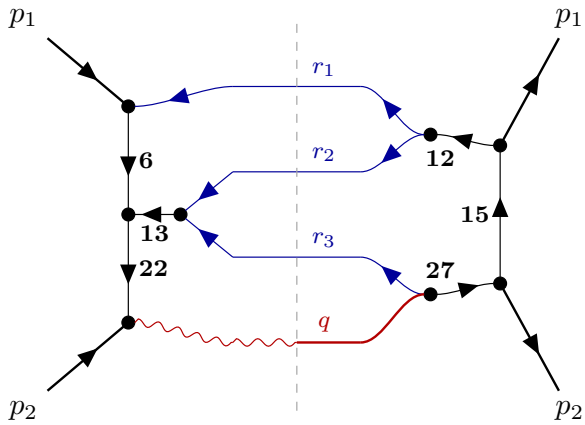
irreducible numerators: $25 = (p_1 + p_2 + r_2)^2$



FAM[640] (topo640)

6	$(p_1 + r_1)^2$	amplitude
13	$(r_2 + r_3)^2$	amplitude
20	$(p_2 + r_2 + r_3)^2$	amplitude
12	$(r_1 + r_2)^2$	conjugate
15	$(p_1 + r_1 + r_2)^2$	conjugate
27	$(p_1 + p_2 + r_1 + r_2)^2$	conjugate

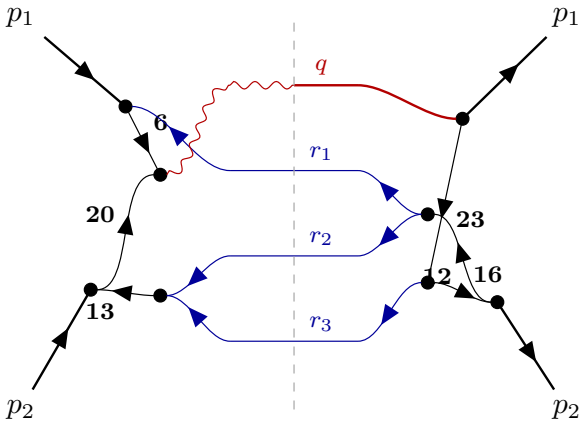
irreducible numerators: $21 = (r_1 + r_2 + r_3)^2$



FAM[646] (topo646)

6	$(p_1 + r_1)^2$	amplitude
13	$(r_2 + r_3)^2$	amplitude
22	$(p_1 + r_1 + r_2 + r_3)^2$	amplitude
12	$(r_1 + r_2)^2$	conjugate
15	$(p_1 + r_1 + r_2)^2$	conjugate
27	$(p_1 + p_2 + r_1 + r_2)^2$	conjugate

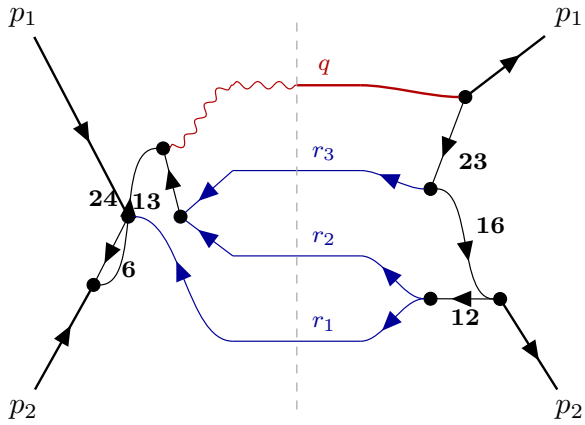
irreducible numerators: $24 = (p_1 + p_2 + r_1)^2$



FAM[652] (topo652)

6	$(p_1 + r_1)^2$	amplitude
13	$(r_2 + r_3)^2$	amplitude
20	$(p_2 + r_2 + r_3)^2$	amplitude
12	$(r_1 + r_2)^2$	conjugate
16	$(p_2 + r_1 + r_2)^2$	conjugate
23	$(p_2 + r_1 + r_2 + r_3)^2$	conjugate

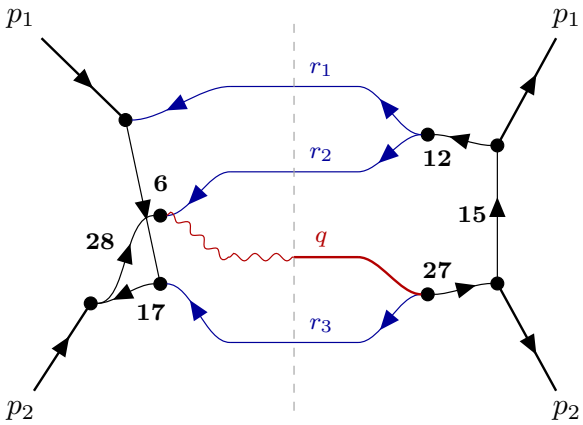
irreducible numerators: $27 = (p_1 + p_2 + r_1 + r_2)^2$



FAM[658] (topo658)

6	$(p_1 + r_1)^2$	amplitude
13	$(r_2 + r_3)^2$	amplitude
24	$(p_1 + p_2 + r_1)^2$	amplitude
12	$(r_1 + r_2)^2$	conjugate
16	$(p_2 + r_1 + r_2)^2$	conjugate
23	$(p_2 + r_1 + r_2 + r_3)^2$	conjugate

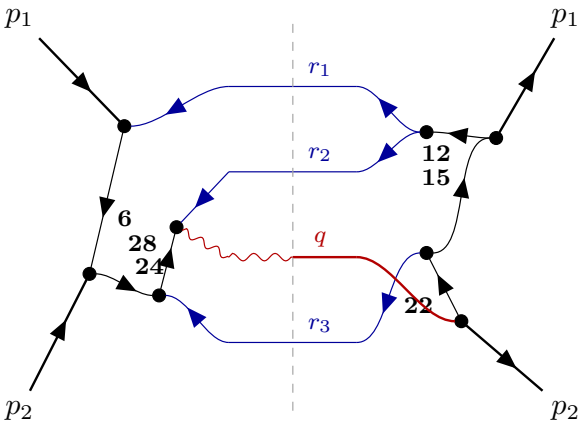
irreducible numerators: $28 = (p_1 + p_2 + r_1 + r_3)^2$



FAM[664] (topo664)

6	$(p_1 + r_1)^2$	amplitude
17	$(p_1 + r_1 + r_3)^2$	amplitude
28	$(p_1 + p_2 + r_1 + r_3)^2$	amplitude
12	$(r_1 + r_2)^2$	conjugate
15	$(p_1 + r_1 + r_2)^2$	conjugate
27	$(p_1 + p_2 + r_1 + r_2)^2$	conjugate

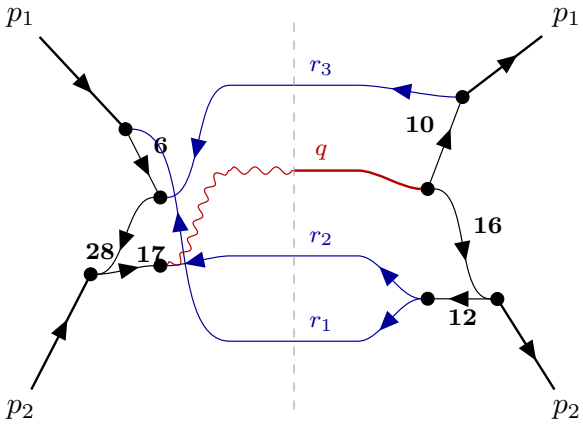
irreducible numerators: $26 = (p_1 + p_2 + r_3)^2$



FAM[670] (topo670)

6	$(p_1 + r_1)^2$	amplitude
24	$(p_1 + p_2 + r_1)^2$	amplitude
28	$(p_1 + p_2 + r_1 + r_3)^2$	amplitude
12	$(r_1 + r_2)^2$	conjugate
15	$(p_1 + r_1 + r_2)^2$	conjugate
22	$(p_1 + r_1 + r_2 + r_3)^2$	conjugate

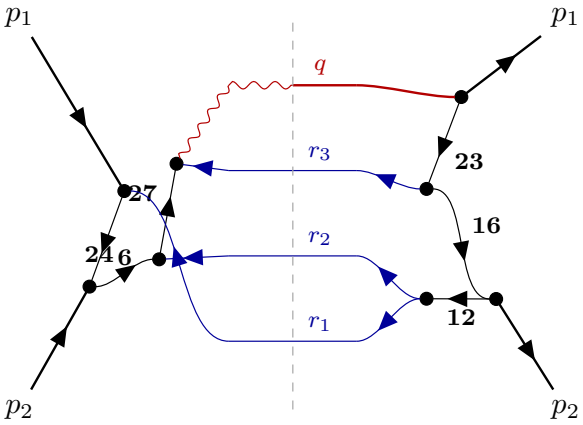
irreducible numerators: $26 = (p_1 + p_2 + r_3)^2$



FAM[676] (topo676)

6	$(p_1 + r_1)^2$	amplitude
17	$(p_1 + r_1 + r_3)^2$	amplitude
28	$(p_1 + p_2 + r_1 + r_3)^2$	amplitude
10	$(p_1 + r_3)^2$	conjugate
12	$(r_1 + r_2)^2$	conjugate
16	$(p_2 + r_1 + r_2)^2$	conjugate

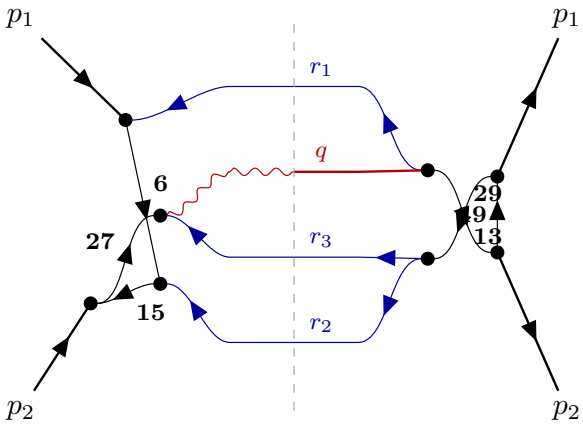
irreducible numerators: $23 = (p_2 + r_1 + r_2 + r_3)^2$



FAM[682] (topo682)

6	$(p_1 + r_1)^2$	amplitude
24	$(p_1 + p_2 + r_1)^2$	amplitude
27	$(p_1 + p_2 + r_1 + r_2)^2$	amplitude
12	$(r_1 + r_2)^2$	conjugate
16	$(p_2 + r_1 + r_2)^2$	conjugate
23	$(p_2 + r_1 + r_2 + r_3)^2$	conjugate

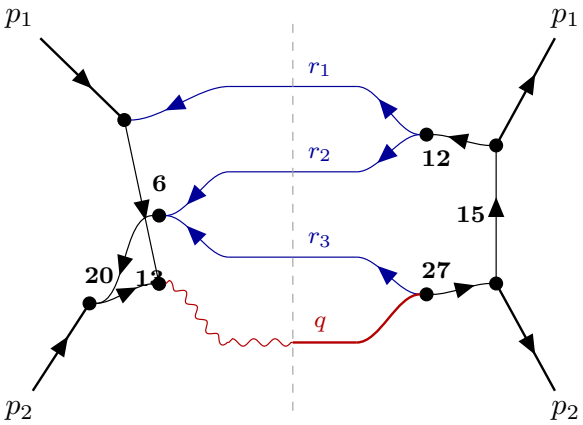
irreducible numerators: $28 = (p_1 + p_2 + r_1 + r_3)^2$



FAM[688] (topo688)

6	$(p_1 + r_1)^2$	amplitude
15	$(p_1 + r_1 + r_2)^2$	amplitude
27	$(p_1 + p_2 + r_1 + r_2)^2$	amplitude
13	$(r_2 + r_3)^2$	conjugate
19	$(p_1 + r_2 + r_3)^2$	conjugate
29	$(p_1 + p_2 + r_2 + r_3)^2$	conjugate

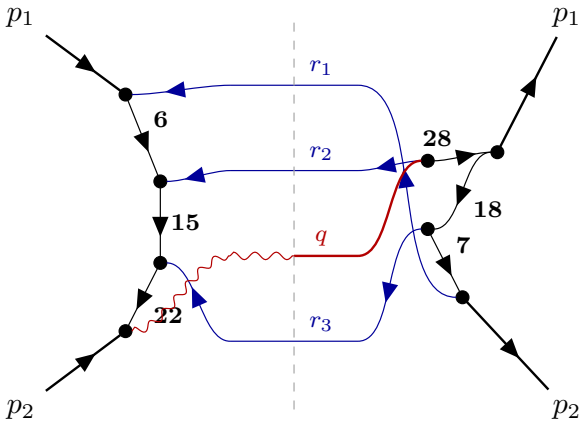
irreducible numerators: $12 = (r_1 + r_2)^2$



FAM[694] (topo694)

6	$(p_1 + r_1)^2$	amplitude
13	$(r_2 + r_3)^2$	amplitude
20	$(p_2 + r_2 + r_3)^2$	amplitude
12	$(r_1 + r_2)^2$	conjugate
15	$(p_1 + r_1 + r_2)^2$	conjugate
27	$(p_1 + p_2 + r_1 + r_2)^2$	conjugate

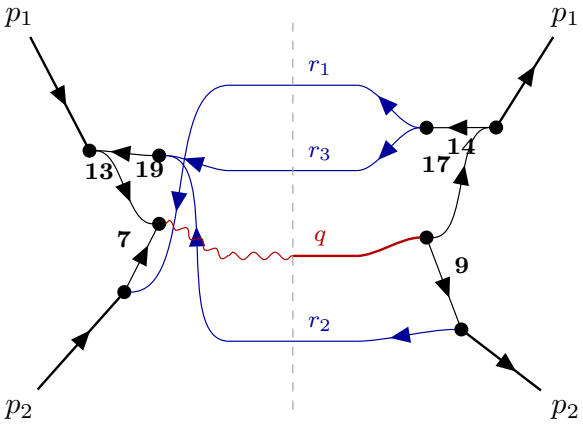
irreducible numerators: $29 = (p_1 + p_2 + r_2 + r_3)^2$



FAM[700] (topo700)

6	$(p_1 + r_1)^2$	amplitude
15	$(p_1 + r_1 + r_2)^2$	amplitude
22	$(p_1 + r_1 + r_2 + r_3)^2$	amplitude
7	$(p_2 + r_1)^2$	conjugate
18	$(p_2 + r_1 + r_3)^2$	conjugate
28	$(p_1 + p_2 + r_1 + r_3)^2$	conjugate

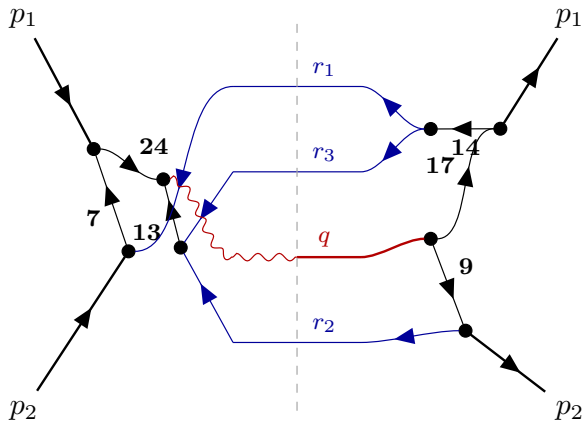
irreducible numerators: $14 = (r_1 + r_3)^2$



FAM[706] (topo706)

7	$(p_2 + r_1)^2$	amplitude
13	$(r_2 + r_3)^2$	amplitude
19	$(p_1 + r_2 + r_3)^2$	amplitude
9	$(p_2 + r_2)^2$	conjugate
14	$(r_1 + r_3)^2$	conjugate
17	$(p_1 + r_1 + r_3)^2$	conjugate

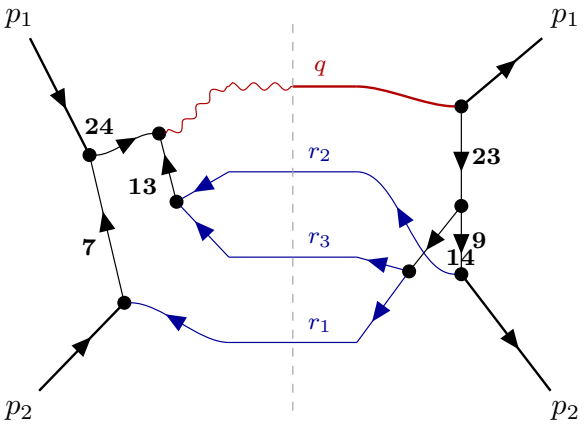
irreducible numerators: $22 = (p_1 + r_1 + r_2 + r_3)^2$



FAM[709] (topo709)

7	$(p_2 + r_1)^2$	amplitude
13	$(r_2 + r_3)^2$	amplitude
24	$(p_1 + p_2 + r_1)^2$	amplitude
9	$(p_2 + r_2)^2$	conjugate
14	$(r_1 + r_3)^2$	conjugate
17	$(p_1 + r_1 + r_3)^2$	conjugate

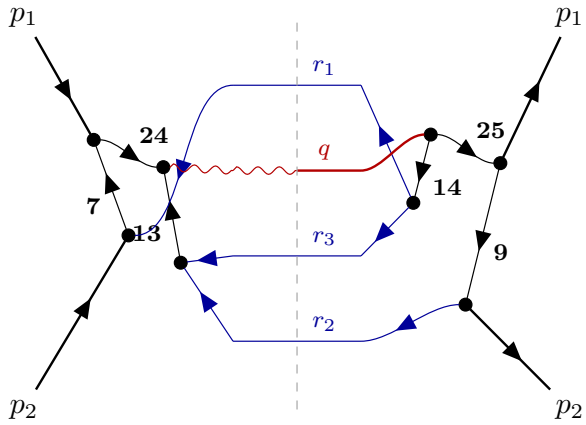
irreducible numerators: $28 = (p_1 + p_2 + r_1 + r_3)^2$



FAM[715] (topo715)

7	$(p_2 + r_1)^2$	amplitude
13	$(r_2 + r_3)^2$	amplitude
24	$(p_1 + p_2 + r_1)^2$	amplitude
9	$(p_2 + r_2)^2$	conjugate
14	$(r_1 + r_3)^2$	conjugate
23	$(p_2 + r_1 + r_2 + r_3)^2$	conjugate

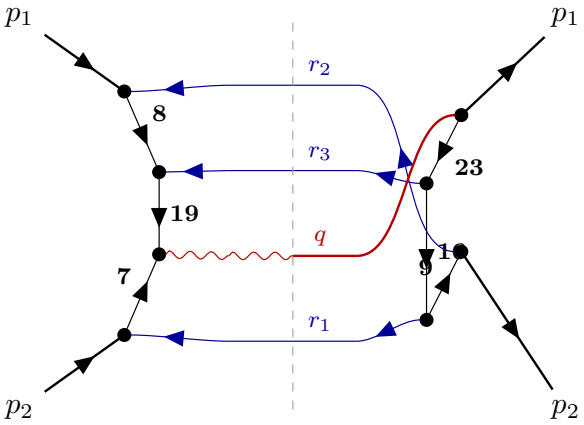
irreducible numerators: $27 = (p_1 + p_2 + r_1 + r_2)^2$



FAM[721] (topo721)

7	$(p_2 + r_1)^2$	amplitude
13	$(r_2 + r_3)^2$	amplitude
24	$(p_1 + p_2 + r_1)^2$	amplitude
9	$(p_2 + r_2)^2$	conjugate
14	$(r_1 + r_3)^2$	conjugate
25	$(p_1 + p_2 + r_2)^2$	conjugate

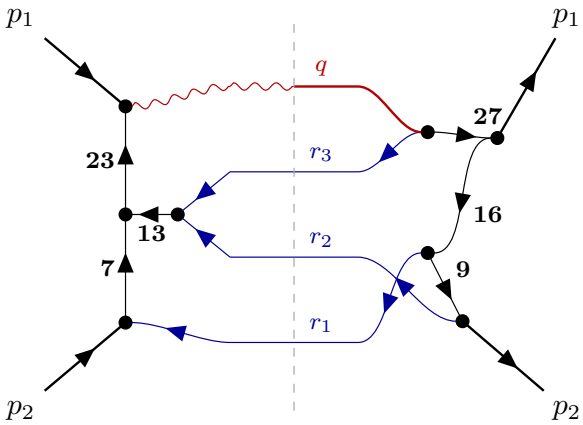
irreducible numerators: $18 = (p_2 + r_1 + r_3)^2$



FAM[727] (topo727)

7	$(p_2 + r_1)^2$	amplitude
8	$(p_1 + r_2)^2$	amplitude
19	$(p_1 + r_2 + r_3)^2$	amplitude
9	$(p_2 + r_2)^2$	conjugate
16	$(p_2 + r_1 + r_2)^2$	conjugate
23	$(p_2 + r_1 + r_2 + r_3)^2$	conjugate

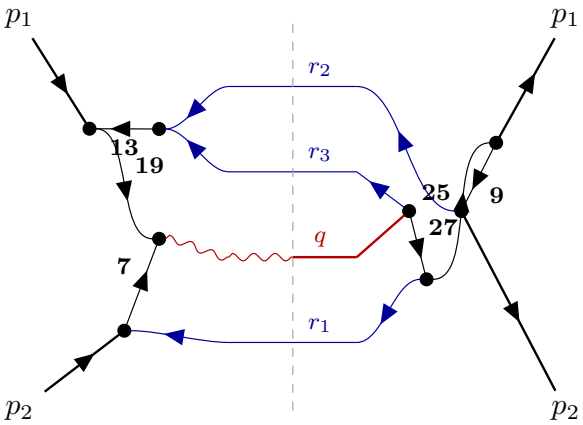
irreducible numerators: $13 = (r_2 + r_3)^2$



FAM[733] (topo733)

7	$(p_2 + r_1)^2$	amplitude
13	$(r_2 + r_3)^2$	amplitude
23	$(p_2 + r_1 + r_2 + r_3)^2$	amplitude
9	$(p_2 + r_2)^2$	conjugate
16	$(p_2 + r_1 + r_2)^2$	conjugate
27	$(p_1 + p_2 + r_1 + r_2)^2$	conjugate

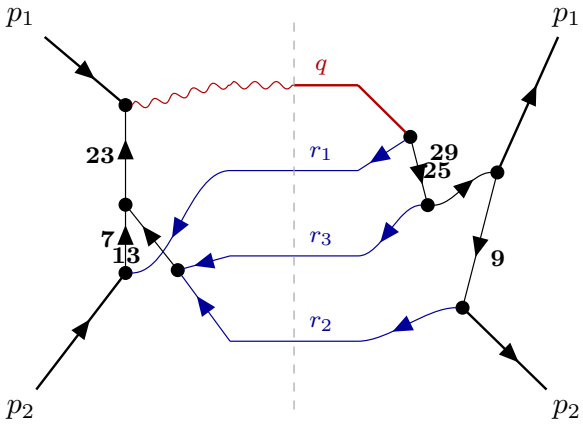
irreducible numerators: $24 = (p_1 + p_2 + r_1)^2$



FAM[739] (topo739)

7	$(p_2 + r_1)^2$	amplitude
13	$(r_2 + r_3)^2$	amplitude
19	$(p_1 + r_2 + r_3)^2$	amplitude
9	$(p_2 + r_2)^2$	conjugate
25	$(p_1 + p_2 + r_2)^2$	conjugate
27	$(p_1 + p_2 + r_1 + r_2)^2$	conjugate

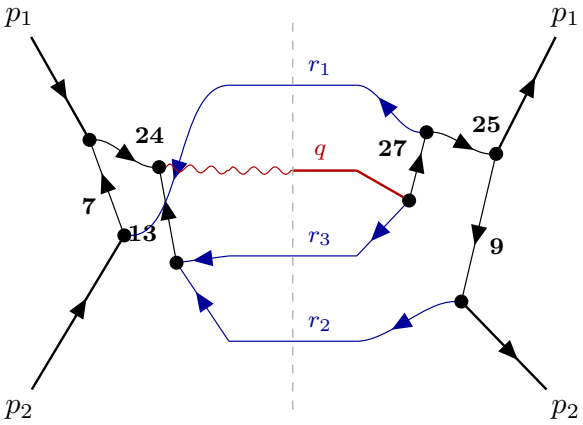
irreducible numerators: $22 = (p_1 + r_1 + r_2 + r_3)^2$



FAM[745] (topo745)

7	$(p_2 + r_1)^2$	amplitude
13	$(r_2 + r_3)^2$	amplitude
23	$(p_2 + r_1 + r_2 + r_3)^2$	amplitude
9	$(p_2 + r_2)^2$	conjugate
25	$(p_1 + p_2 + r_2)^2$	conjugate
29	$(p_1 + p_2 + r_2 + r_3)^2$	conjugate

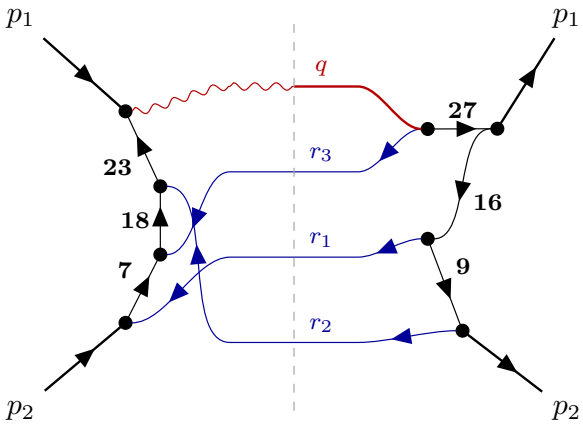
irreducible numerators: $12 = (r_1 + r_2)^2$



FAM[751] (topo751)

7	$(p_2 + r_1)^2$	amplitude
13	$(r_2 + r_3)^2$	amplitude
24	$(p_1 + p_2 + r_1)^2$	amplitude
9	$(p_2 + r_2)^2$	conjugate
25	$(p_1 + p_2 + r_2)^2$	conjugate
27	$(p_1 + p_2 + r_1 + r_2)^2$	conjugate

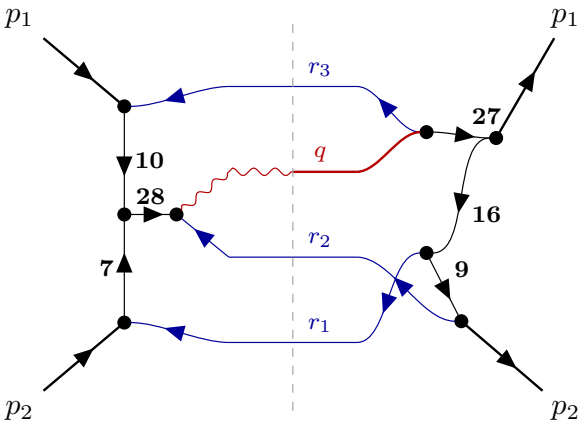
irreducible numerators: $29 = (p_1 + p_2 + r_2 + r_3)^2$



FAM[757] (topo757)

7	$(p_2 + r_1)^2$	amplitude
18	$(p_2 + r_1 + r_3)^2$	amplitude
23	$(p_2 + r_1 + r_2 + r_3)^2$	amplitude
9	$(p_2 + r_2)^2$	conjugate
16	$(p_2 + r_1 + r_2)^2$	conjugate
27	$(p_1 + p_2 + r_1 + r_2)^2$	conjugate

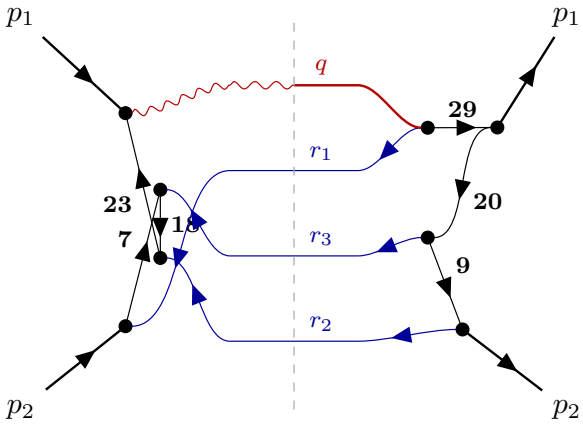
irreducible numerators: $28 = (p_1 + p_2 + r_1 + r_3)^2$



FAM[763] (topo763)

7	$(p_2 + r_1)^2$	amplitude
10	$(p_1 + r_3)^2$	amplitude
28	$(p_1 + p_2 + r_1 + r_3)^2$	amplitude
9	$(p_2 + r_2)^2$	conjugate
16	$(p_2 + r_1 + r_2)^2$	conjugate
27	$(p_1 + p_2 + r_1 + r_2)^2$	conjugate

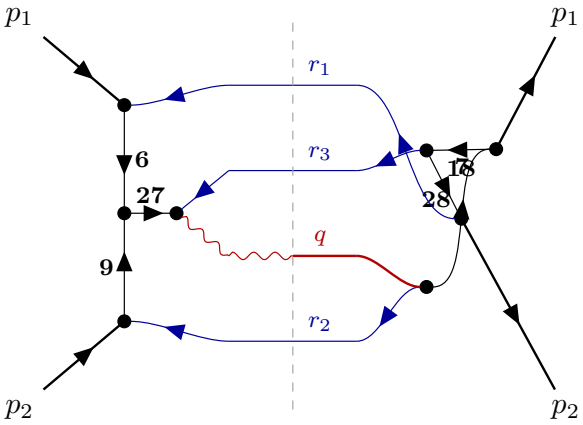
irreducible numerators: $24 = (p_1 + p_2 + r_1)^2$



FAM[769] (topo769)

7	$(p_2 + r_1)^2$	amplitude
18	$(p_2 + r_1 + r_3)^2$	amplitude
23	$(p_2 + r_1 + r_2 + r_3)^2$	amplitude
9	$(p_2 + r_2)^2$	conjugate
20	$(p_2 + r_2 + r_3)^2$	conjugate
29	$(p_1 + p_2 + r_2 + r_3)^2$	conjugate

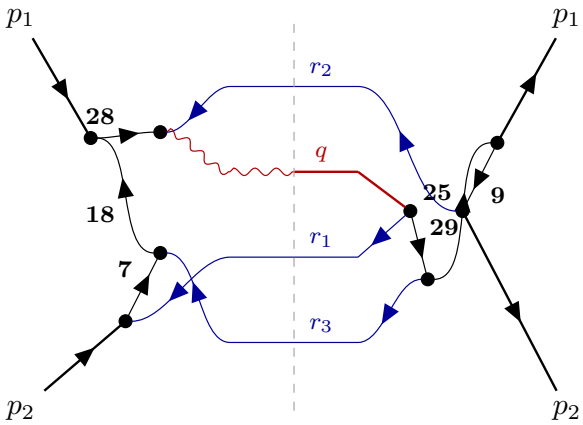
irreducible numerators: $28 = (p_1 + p_2 + r_1 + r_3)^2$



FAM[772] (topo772)

6	$(p_1 + r_1)^2$	amplitude
9	$(p_2 + r_2)^2$	amplitude
27	$(p_1 + p_2 + r_1 + r_2)^2$	amplitude
7	$(p_2 + r_1)^2$	conjugate
18	$(p_2 + r_1 + r_3)^2$	conjugate
28	$(p_1 + p_2 + r_1 + r_3)^2$	conjugate

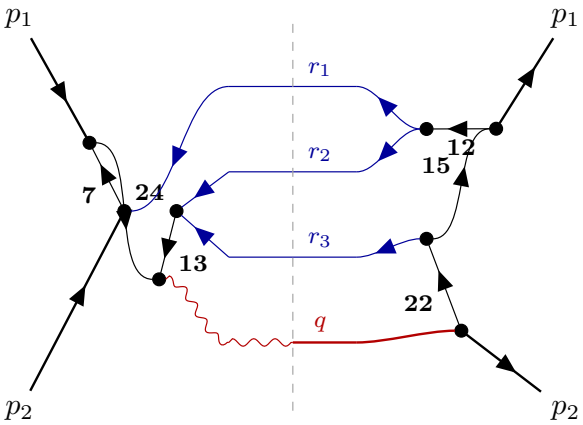
irreducible numerators: $25 = (p_1 + p_2 + r_2)^2$



FAM[778] (topo778)

7	$(p_2 + r_1)^2$	amplitude
18	$(p_2 + r_1 + r_3)^2$	amplitude
28	$(p_1 + p_2 + r_1 + r_3)^2$	amplitude
9	$(p_2 + r_2)^2$	conjugate
25	$(p_1 + p_2 + r_2)^2$	conjugate
29	$(p_1 + p_2 + r_2 + r_3)^2$	conjugate

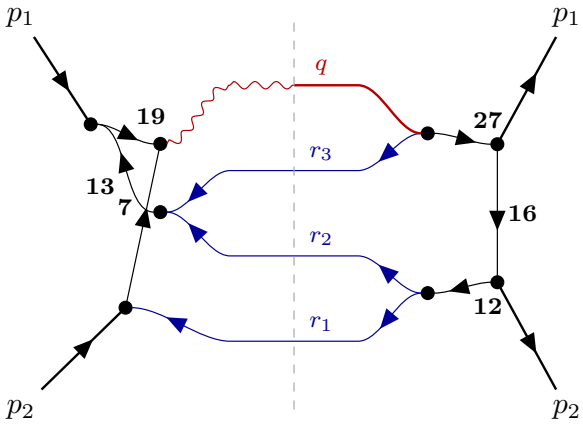
irreducible numerators: $24 = (p_1 + p_2 + r_1)^2$



FAM[784] (topo784)

7	$(p_2 + r_1)^2$	amplitude
13	$(r_2 + r_3)^2$	amplitude
24	$(p_1 + p_2 + r_1)^2$	amplitude
12	$(r_1 + r_2)^2$	conjugate
15	$(p_1 + r_1 + r_2)^2$	conjugate
22	$(p_1 + r_1 + r_2 + r_3)^2$	conjugate

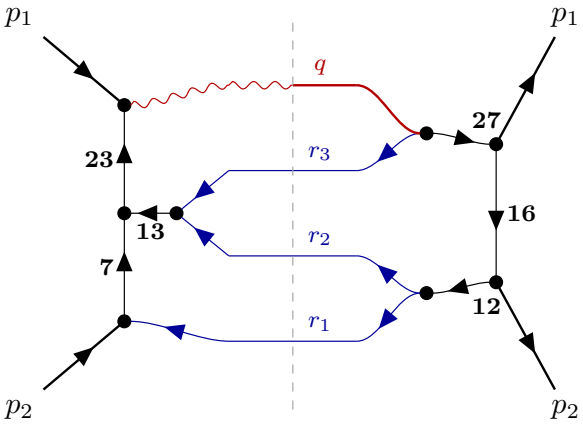
irreducible numerators: $27 = (p_1 + p_2 + r_1 + r_2)^2$



FAM[790] (topo790)

7	$(p_2 + r_1)^2$	amplitude
13	$(r_2 + r_3)^2$	amplitude
19	$(p_1 + r_2 + r_3)^2$	amplitude
12	$(r_1 + r_2)^2$	conjugate
16	$(p_2 + r_1 + r_2)^2$	conjugate
27	$(p_1 + p_2 + r_1 + r_2)^2$	conjugate

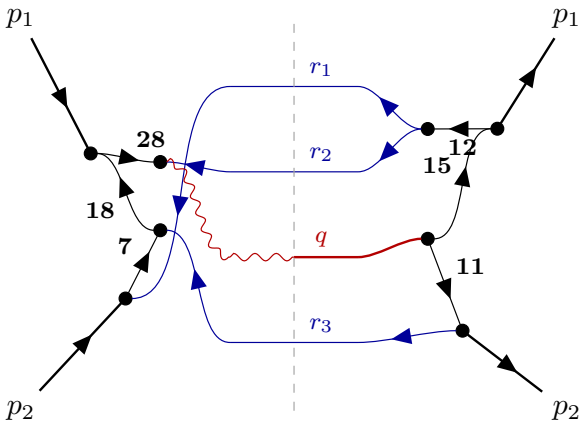
irreducible numerators: $21 = (r_1 + r_2 + r_3)^2$



FAM[796] (topo796)

7	$(p_2 + r_1)^2$	amplitude
13	$(r_2 + r_3)^2$	amplitude
23	$(p_2 + r_1 + r_2 + r_3)^2$	amplitude
12	$(r_1 + r_2)^2$	conjugate
16	$(p_2 + r_1 + r_2)^2$	conjugate
27	$(p_1 + p_2 + r_1 + r_2)^2$	conjugate

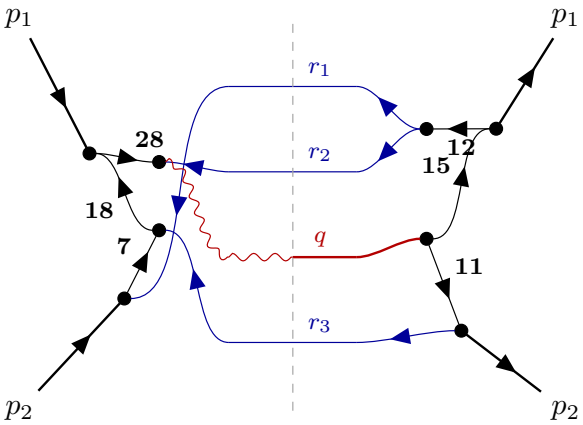
irreducible numerators: $24 = (p_1 + p_2 + r_1)^2$



FAM[802] (topo802)

7	$(p_2 + r_1)^2$	amplitude
18	$(p_2 + r_1 + r_3)^2$	amplitude
28	$(p_1 + p_2 + r_1 + r_3)^2$	amplitude
11	$(p_2 + r_3)^2$	conjugate
12	$(r_1 + r_2)^2$	conjugate
15	$(p_1 + r_1 + r_2)^2$	conjugate

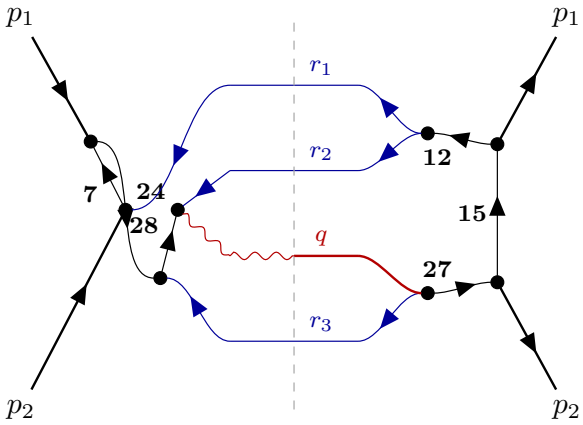
irreducible numerators: $22 = (p_1 + r_1 + r_2 + r_3)^2$



FAM[808] (topo808)

7	$(p_2 + r_1)^2$	amplitude
18	$(p_2 + r_1 + r_3)^2$	amplitude
28	$(p_1 + p_2 + r_1 + r_3)^2$	amplitude
11	$(p_2 + r_3)^2$	conjugate
12	$(r_1 + r_2)^2$	conjugate
15	$(p_1 + r_1 + r_2)^2$	conjugate

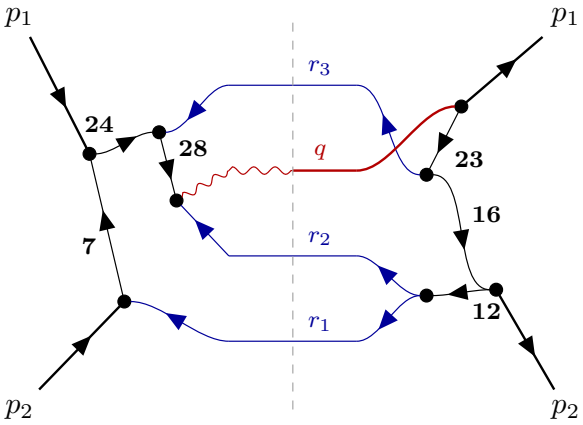
irreducible numerators: $27 = (p_1 + p_2 + r_1 + r_2)^2$



FAM[814] (topo814)

7	$(p_2 + r_1)^2$	amplitude
24	$(p_1 + p_2 + r_1)^2$	amplitude
28	$(p_1 + p_2 + r_1 + r_3)^2$	amplitude
12	$(r_1 + r_2)^2$	conjugate
15	$(p_1 + r_1 + r_2)^2$	conjugate
27	$(p_1 + p_2 + r_1 + r_2)^2$	conjugate

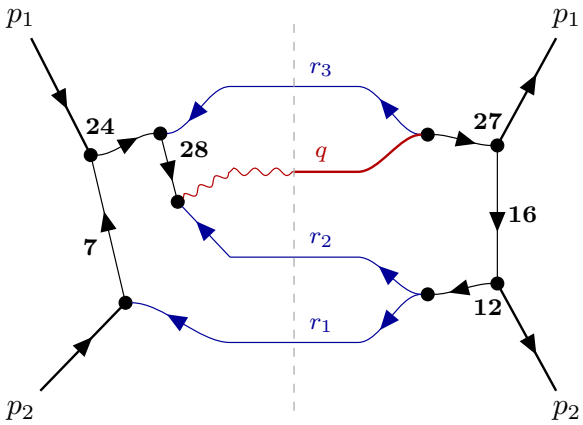
irreducible numerators: $21 = (r_1 + r_2 + r_3)^2$



FAM[820] (topo820)

7	$(p_2 + r_1)^2$	amplitude
24	$(p_1 + p_2 + r_1)^2$	amplitude
28	$(p_1 + p_2 + r_1 + r_3)^2$	amplitude
12	$(r_1 + r_2)^2$	conjugate
16	$(p_2 + r_1 + r_2)^2$	conjugate
23	$(p_2 + r_1 + r_2 + r_3)^2$	conjugate

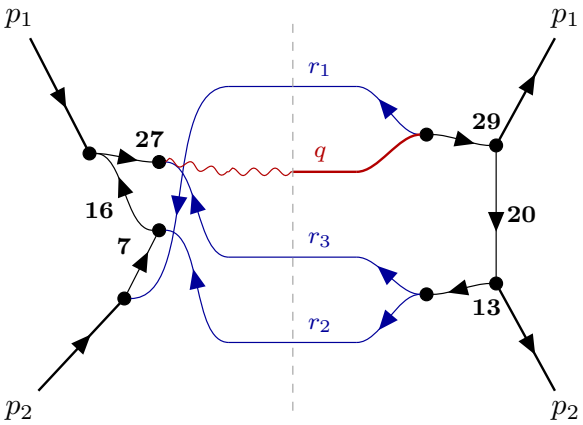
irreducible numerators: $26 = (p_1 + p_2 + r_3)^2$



FAM[826] (topo826)

7	$(p_2 + r_1)^2$	amplitude
24	$(p_1 + p_2 + r_1)^2$	amplitude
28	$(p_1 + p_2 + r_1 + r_3)^2$	amplitude
12	$(r_1 + r_2)^2$	conjugate
16	$(p_2 + r_1 + r_2)^2$	conjugate
27	$(p_1 + p_2 + r_1 + r_2)^2$	conjugate

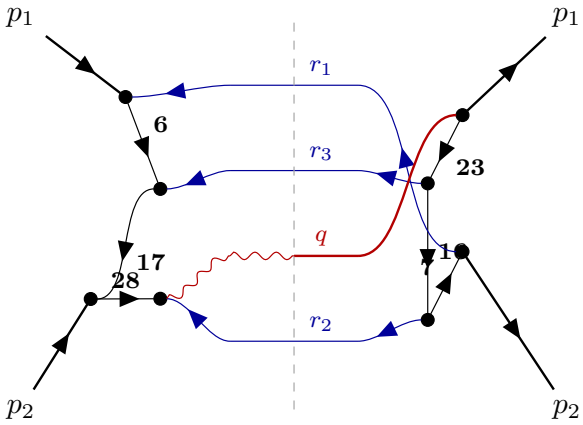
irreducible numerators: $26 = (p_1 + p_2 + r_3)^2$



FAM[832] (topo832)

7	$(p_2 + r_1)^2$	amplitude
16	$(p_2 + r_1 + r_2)^2$	amplitude
27	$(p_1 + p_2 + r_1 + r_2)^2$	amplitude
13	$(r_2 + r_3)^2$	conjugate
20	$(p_2 + r_2 + r_3)^2$	conjugate
29	$(p_1 + p_2 + r_2 + r_3)^2$	conjugate

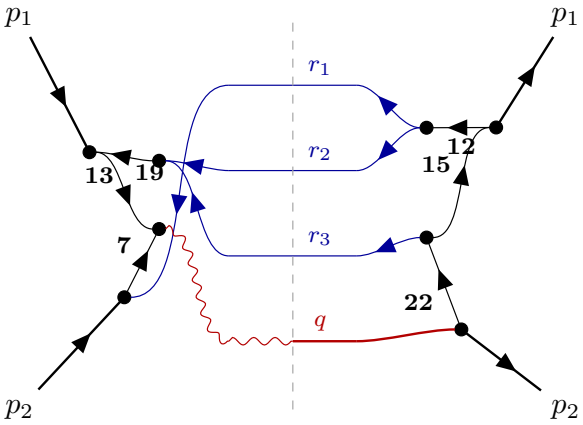
irreducible numerators: $12 = (r_1 + r_2)^2$



FAM[838] (topo838)

6	$(p_1 + r_1)^2$	amplitude
17	$(p_1 + r_1 + r_3)^2$	amplitude
28	$(p_1 + p_2 + r_1 + r_3)^2$	amplitude
7	$(p_2 + r_1)^2$	conjugate
16	$(p_2 + r_1 + r_2)^2$	conjugate
23	$(p_2 + r_1 + r_2 + r_3)^2$	conjugate

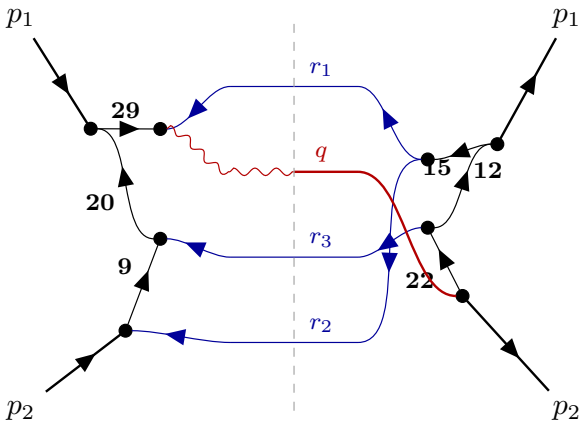
irreducible numerators: $14 = (r_1 + r_3)^2$



FAM[844] (topo844)

7	$(p_2 + r_1)^2$	amplitude
13	$(r_2 + r_3)^2$	amplitude
19	$(p_1 + r_2 + r_3)^2$	amplitude
12	$(r_1 + r_2)^2$	conjugate
15	$(p_1 + r_1 + r_2)^2$	conjugate
22	$(p_1 + r_1 + r_2 + r_3)^2$	conjugate

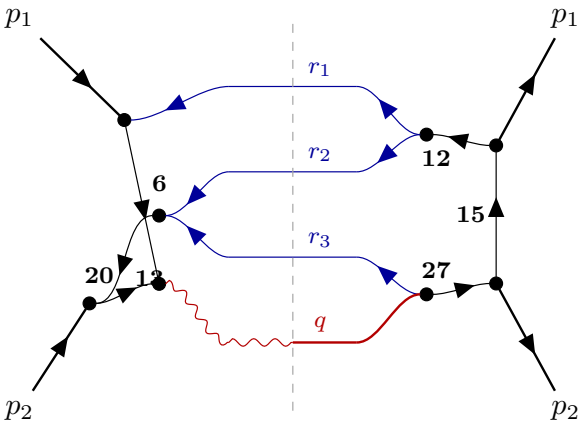
irreducible numerators: $27 = (p_1 + p_2 + r_1 + r_2)^2$



FAM[850] (topo850)

9	$(p_2 + r_2)^2$	amplitude
20	$(p_2 + r_2 + r_3)^2$	amplitude
29	$(p_1 + p_2 + r_2 + r_3)^2$	amplitude
12	$(r_1 + r_2)^2$	conjugate
15	$(p_1 + r_1 + r_2)^2$	conjugate
22	$(p_1 + r_1 + r_2 + r_3)^2$	conjugate

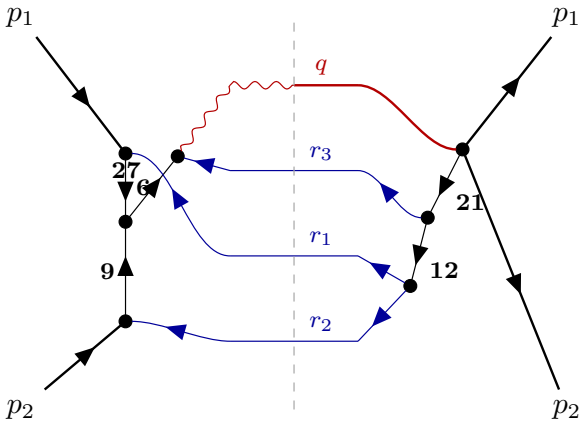
irreducible numerators: $13 = (r_2 + r_3)^2$



FAM[856] (topo856)

6	$(p_1 + r_1)^2$	amplitude
13	$(r_2 + r_3)^2$	amplitude
20	$(p_2 + r_2 + r_3)^2$	amplitude
12	$(r_1 + r_2)^2$	conjugate
15	$(p_1 + r_1 + r_2)^2$	conjugate
27	$(p_1 + p_2 + r_1 + r_2)^2$	conjugate

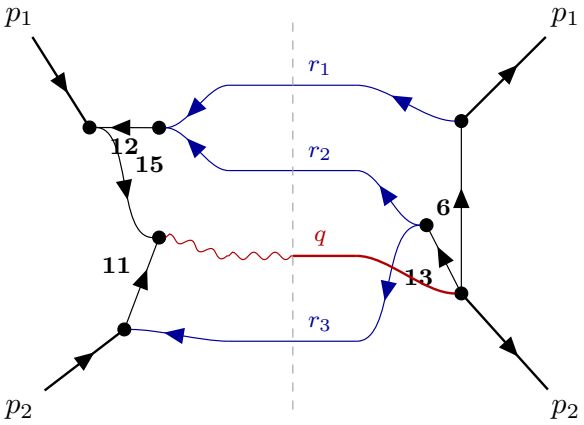
irreducible numerators: $23 = (p_2 + r_1 + r_2 + r_3)^2$



FAM[862] (topo862)

6	$(p_1 + r_1)^2$	amplitude
9	$(p_2 + r_2)^2$	amplitude
27	$(p_1 + p_2 + r_1 + r_2)^2$	amplitude
12	$(r_1 + r_2)^2$	conjugate
21	$(r_1 + r_2 + r_3)^2$	conjugate

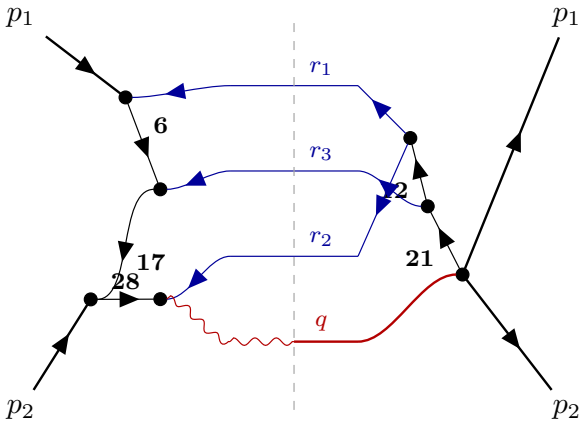
irreducible numerators: $14 = (r_1 + r_3)^2$, $24 = (p_1 + p_2 + r_1)^2$



FAM[868] (topo868)

11	$(p_2 + r_3)^2$	amplitude
12	$(r_1 + r_2)^2$	amplitude
15	$(p_1 + r_1 + r_2)^2$	amplitude
6	$(p_1 + r_1)^2$	conjugate
13	$(r_2 + r_3)^2$	conjugate

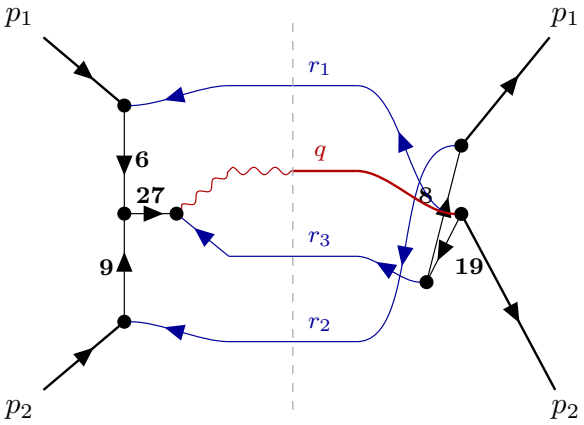
irreducible numerators: $7 = (p_2 + r_1)^2$, $26 = (p_1 + p_2 + r_3)^2$



FAM[874] (topo874)

6	$(p_1 + r_1)^2$	amplitude
17	$(p_1 + r_1 + r_3)^2$	amplitude
28	$(p_1 + p_2 + r_1 + r_3)^2$	amplitude
12	$(r_1 + r_2)^2$	conjugate
21	$(r_1 + r_2 + r_3)^2$	conjugate

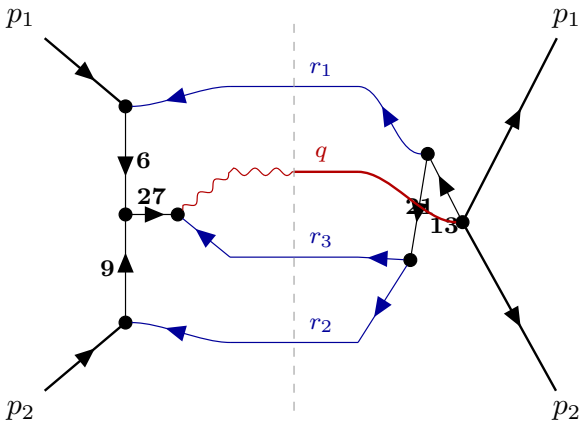
irreducible numerators: $7 = (p_2 + r_1)^2$, $18 = (p_2 + r_1 + r_3)^2$



FAM[880] (topo880)

6	$(p_1 + r_1)^2$	amplitude
9	$(p_2 + r_2)^2$	amplitude
27	$(p_1 + p_2 + r_1 + r_2)^2$	amplitude
8	$(p_1 + r_2)^2$	conjugate
19	$(p_1 + r_2 + r_3)^2$	conjugate

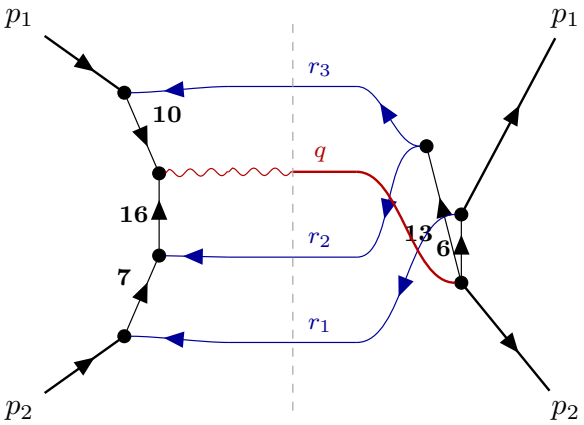
irreducible numerators: $13 = (r_2 + r_3)^2$, $24 = (p_1 + p_2 + r_1)^2$



FAM[886] (topo886)

6	$(p_1 + r_1)^2$	amplitude
9	$(p_2 + r_2)^2$	amplitude
27	$(p_1 + p_2 + r_1 + r_2)^2$	amplitude
13	$(r_2 + r_3)^2$	conjugate
21	$(r_1 + r_2 + r_3)^2$	conjugate

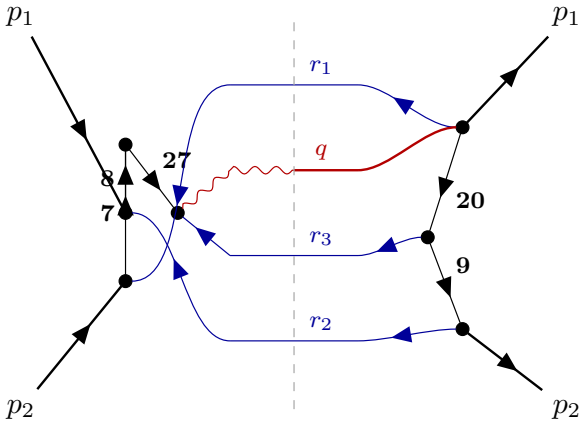
irreducible numerators: $8 = (p_1 + r_2)^2$, $24 = (p_1 + p_2 + r_1)^2$



FAM[892] (topo892)

7	$(p_2 + r_1)^2$	amplitude
10	$(p_1 + r_3)^2$	amplitude
16	$(p_2 + r_1 + r_2)^2$	amplitude
6	$(p_1 + r_1)^2$	conjugate
13	$(r_2 + r_3)^2$	conjugate

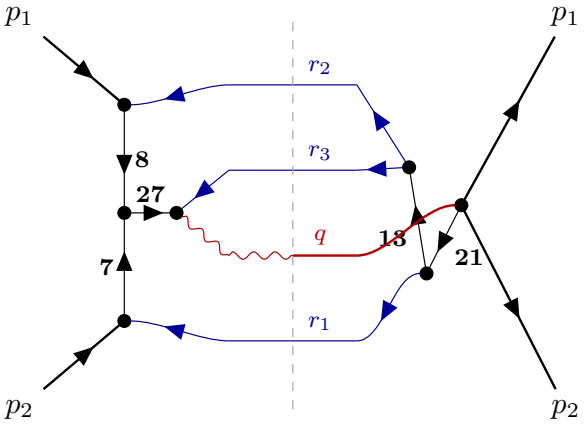
irreducible numerators: $12 = (r_1 + r_2)^2$, $26 = (p_1 + p_2 + r_3)^2$



FAM[898] (topo898)

7	$(p_2 + r_1)^2$	amplitude
8	$(p_1 + r_2)^2$	amplitude
27	$(p_1 + p_2 + r_1 + r_2)^2$	amplitude
9	$(p_2 + r_2)^2$	conjugate
20	$(p_2 + r_2 + r_3)^2$	conjugate

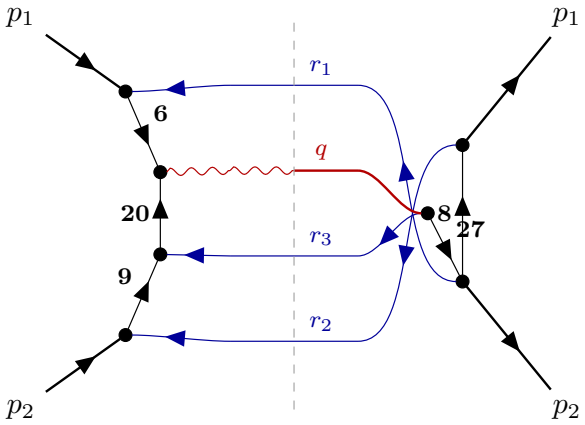
irreducible numerators: $13 = (r_2 + r_3)^2$, $24 = (p_1 + p_2 + r_1)^2$



FAM[904] (topo904)

7	$(p_2 + r_1)^2$	amplitude
8	$(p_1 + r_2)^2$	amplitude
27	$(p_1 + p_2 + r_1 + r_2)^2$	amplitude
13	$(r_2 + r_3)^2$	conjugate
21	$(r_1 + r_2 + r_3)^2$	conjugate

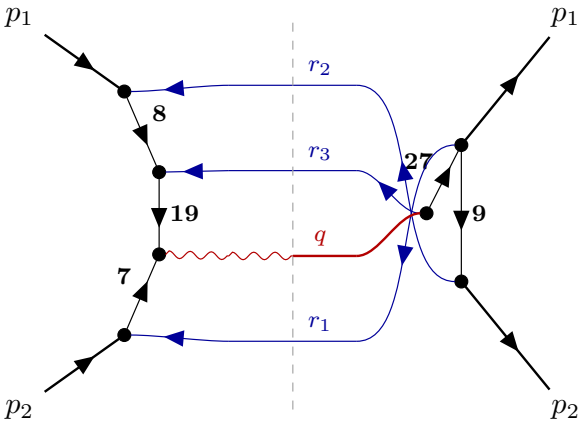
irreducible numerators: $9 = (p_2 + r_2)^2$, $24 = (p_1 + p_2 + r_1)^2$



FAM[910] (topo910)

6	$(p_1 + r_1)^2$	amplitude
9	$(p_2 + r_2)^2$	amplitude
20	$(p_2 + r_2 + r_3)^2$	amplitude
8	$(p_1 + r_2)^2$	conjugate
27	$(p_1 + p_2 + r_1 + r_2)^2$	conjugate

irreducible numerators: $13 = (r_2 + r_3)^2$, $24 = (p_1 + p_2 + r_1)^2$



FAM[916] (topo916)

7	$(p_2 + r_1)^2$	amplitude
8	$(p_1 + r_2)^2$	amplitude
19	$(p_1 + r_2 + r_3)^2$	amplitude
9	$(p_2 + r_2)^2$	conjugate
27	$(p_1 + p_2 + r_1 + r_2)^2$	conjugate

irreducible numerators: $13 = (r_2 + r_3)^2$, $24 = (p_1 + p_2 + r_1)^2$